How do familial demographics influence student performance?

Process Book

Vizzards

Yash Agrawal, Abdulaziz Alajlan, Joshua Blackmon, Justin Bimal, Timi Omoteso

Team Members

Abdulaziz Alajlan (Team leader) aalajlan6@gatech.edu

Josh Blackmon Jblackmon9@gatech.edu

Yash Agrawal Yagrawal36@gatech.edu

Timi Omoteso Oomoteso3@gatech.edu

Justin Bimal Jbimal3@gatech.edu

Team Contract

See the signed contract below

- Each week our team will meet online virtually through a Teams call or Phone call for at least 15 minutes on Tuesday 6:30 pm evenings.
- These meetings will assign each team member tasks for each week and what they will need to get done/report on/put on our Teams Channel by next week
- We will use Teams for communication, and we expect members to respond to pings within a few days unless exceptional circumstances.
- Work is done online, individually, and remotely, but can be on call through separate meetings from our weekly meeting as needed. We can have working sessions between multiple members. These separate meetings will either be notified through pings or be discussed in the weekly meeting. Each member will push any changes onto the Git or put their work onto the Teams channel.
- We will use Git to help combine our work and make sure that no one's work is lost
- Work hours should be split evenly where no team member should be putting in too much work or too little. As part of the weekly meeting, we discuss what tasks we each want to do and ensure that we unanimously agree to a plan for the following week. We will keep each other accountable. If we think anyone is not being accountable for their work, we will discuss so during the weekly meetings and hopefully find a good compromise, and if it continues to a point that we cannot, we will try to get the TA's help.

Signatures: Yash Agrawal, Abdulaziz Alajlan, Joshua Blackmon, Justin Bimal, Timi Omoteso

Date: 9/3/2026

Initial Planning

Ideas Brainstorming

We looked at a variety of datasets before we determined what would be our total and complete project idea. Some of the datasets we looked at include:

- Plane Bird strike data, possibly looking at an interactive US map of the data
    - https://www.kaggle.com/datasets/iamtapendu/bird-strike-by-aircafts-data
- Student Performance data based on parental data
    - Student Performance Dataset | Kaggle
- LLM Benchmarks for Voice Agent data comparing quality, TTFT, LLM-as-a-judge scoring

After looking at all the ideas, we determined Student Performance data based on parental data should be our project idea.

Project

Abstract

Our project will be an exploratory data analysis with the goal of discovering how certain student demographic data may be used to predict academic performance. We will achieve this through visualizing data such as student age, family background, and even habits such as eating out regularly or attending college prep courses. The datasets we chose have a few different ways of defining academic performance, but they are mostly test scores and general numeric grades. Some of our datasets are very dense and have numerous features, while others have as little as two total features. We hope that by looking through numerous datasets, we can find some general trends that can be useful predictors of student success. The intended audience can be students, parents, educators, and even school boards looking to determine what student habits may be negatively or positively impacting grades. Because our data is a mix of nominal and numerical, we will likely have various forms of charts that will be easily understandable to the general public. Overall, we hope to analyse the topic of student performance through the lens of factors both in and out of their own control and determine what factors may contribute the most influence to academic achievement.

Datasets

https://archive.ics.uci.edu/dataset/320/student+performance

https://www.kaggle.com/datasets/shilton123456/eda-on-students-performance-in-subjects

https://www.kaggle.com/datasets/sujaykapadnis/student-grades-prediction-dataset

https://www.kaggle.com/datasets/kuchhbhi/cpga-iq-placement

https://www.kaggle.com/datasets/mahmoudelhemaly/students-grading-dataset

CS 6730

Data Visualization
Map & Storyboard

Team members
Abdulaziz Alajlan
Yash Agrawal
Justin Bimal
Joshua Blackmon
Oluwatimi Omoteso

1. Data Collection and Project Plan

As a team, you need to discuss the following questions and document them in your process book.

I. Who is your audience? Come up with at least three options and pick one target audience.

- Students
- Parents
- Educators
- School Board – Our Target

II. Describe your target audience in more detail. What do they know? What are their interests? What visualization literacy do they have? At what level of detail will you present information to them?

Our target audience is a high school board of education who have a general knowledge of the workings of the school and educators. They also likely know the general makeup and demographics of the school through survey and census data. Their goals should be identifying what factors both in and out of student control that may be influencing student grades, in order to enact policies to improve the overall grades of the school system. As professionals on a school board, they can likely understand and interpret a wide array of visualizations, but we want to keep the depth about what would be seen as a presentation given to the school board. Information will likely be presented at a surface level, showing how certain variables may have a small or large effect on overall grades. Information can be grouped into general categories such as student habits, student living situations, and family history, which can then be used in visualizations to see which of these groupings overall has a larger correlation. Statistics can be used in some visualizations to give context to our data and how it correlates to the entire body of students.

III. What questions about your data will be interesting for your audience? Come up with a list of interesting questions that your audience may have about your data. The more, the better, but your team should come up with at least 15 questions.

1. Which factor that's controllable by the student (such as extra paid classes) leads to the generally most increase in student's grades?
2. Is there a relationship between alcohol consumption and grades for these students?
3. Does familial size correlate to grades? Possibly because the more family members you have, the more time they have to dedicate to their families, but also possibly they have more help from their family?
4. Does wanting to pursue higher education result in higher grades?
5. Are internet access and the scale of going out related? Do students without internet access go out more?
6. Is there an inverse relationship between study time and travel time?
7. Does having gone to nursery school matter in terms of grades? Possibly due to confounding factors with parent wealth? Maybe a stratified analysis would give insight.
8. Does relationship status have a correlation with grades?
9. Are family size and family relationship correlated?
10. Whose education status matters more for the purposes of grades? Mom or dad? Is it by a significant margin?
11. Do absences correlate with lower grades throughout the year?
12. Does student internet access at home have a correlation with grades?
13. Does the guardian status matter for grades?

14. What school has a higher “home” preference rate? How about for “course” or “reputation”? Is that relevant for grades?
15. Which of family support for education and school support has a greater impact on student grades?

IV. What data do you have? Give a brief description of each attribute and its data type (categorical, ordinal, or quantitative) in your process book. It’s OK if you are not sure about the data type for some attributes - you can simply describe them (e.g., geographic location).

Came from https://archive.ics.uci.edu/dataset/320/student+performance variables table

Role	Type	Description	Missing Values
school	Categorical	student's school (binary: 'GP' - Gabriel Pereira or 'MS' - Mousinho da Silveira)	no
sex	Categorical	student's sex (binary: 'F' - female or 'M' - male)	no
age	Quantitative	student's age (numeric: from 15 to 22)	no
address	Categorical	student's home address type (binary: 'U' - urban or 'R' - rural)	no
famsize	Categorical	family size (binary: 'LE3' - less or equal to 3 or 'GT3' - greater than 3)	no
Pstatus	Categorical	parent's cohabitation status (binary: 'T' - living together or 'A' - apart)	no
Medu	Categorical/Ordinal	mother's education (numeric: 0 - none, 1 - primary education (4th grade), 2 - 5th to 9th grade, 3 - secondary education or 4 - higher education)	no
Fedu	Ordinal	father's education (numeric: 0 - none, 1 - primary education (4th grade), 2 – 5th to 9th grade, 3 – secondary education or 4 – higher education)	no
Mjob	Categorical	mother's job (nominal: 'teacher', 'health' care related, civil 'services' (e.g. administrative or police), 'at_home' or 'other')	no
Fjob	Categorical	father's job (nominal: 'teacher', 'health' care related, civil 'services' (e.g. administrative or police), 'at_home' or 'other')	no
reason	Categorical	reason to choose this school (nominal: close to 'home', school 'reputation', 'course' preference or 'other')	no

guardian	Categorical	student's guardian (nominal: 'mother', 'father' or 'other')	no
traveltime	Ordinal/Quantitative	home to school travel time (numeric: 1 - <15 min., 2 - 15 to 30 min., 3 - 30 min. to 1 hour, or 4 - >1 hour)	no
studytime	Categorical/Ordinal	weekly study time (numeric: 1 - <2 hours, 2 - 2 to 5 hours, 3 - 5 to 10 hours, or 4 - >10 hours)	no
failures	Categorical/Ordinal	number of past class failures (numeric: n if 1<=n<3, else 4)	no
schoolsup	Categorical	extra educational support (binary: yes or no)	no
famsup	Categorical	family educational support (binary: yes or no)	no
paid	Categorical	extra paid classes within the course subject (Math or Portuguese) (binary: yes or no)	no
activities	Categorical	extra-curricular activities (binary: yes or no)	no
nursery	Categorical	attended nursery school (binary: yes or no)	no
higher	Categorical	wants to take higher education (binary: yes or no)	no
internet	Categorical	Internet access at home (binary: yes or no)	no
romantic	Categorical	with a romantic relationship (binary: yes or no)	no
famrel	Categorical/Ordinal	quality of family relationships (numeric: from 1 - very bad to 5 - excellent)	no
freetime	Categorical/Ordinal	free time after school (numeric: from 1 - very low to 5 - very high)	no
goout	Categorical	going out with friends (numeric: from 1 - very low to 5 - very high)	no
Dalc	Categorical/Ordinal	workday alcohol consumption (numeric: from 1 - very low to 5 - very high)	no
Walc	Categorical/Ordinal	weekend alcohol consumption (numeric: from 1 - very low to 5 - very high)	no
health	Categorical/Ordinal	current health status (numeric: from 1 - very bad to 5 - very good)	no

absences	Ordinal/Quantitative	number of school absences (numeric: from 0 to 93)	no
G1	Categorical	first period grade (numeric: from 0 to 20)	no
G2	Categorical	second period grade (numeric: from 0 to 20)	no
G3	Quantitative	final grade (numeric: from 0 to 20, output target)	no

V. Do you expect to do substantial data cleanup? How will data processing be implemented? Try to minimize the amount of cleanup you must do by finding cleaned and ready-to-go data sources whenever possible.

Our main dataset has no missing values and has very consistent labels for any categorical data. Most of our data processing will determine whether categories that have vague values (such as “other”) as a majority will be included. These categories may be less useful in interpreting the data as “other” is a wide range of data, and if the majority of an attribute has that value, the attribute may be less useful in visualizing.

2. Data Exploration

You now need to identify the key message and insights from the data.

1. Use Tableau/Excel/Python/R (or any visualization tool you like) to create preliminary visualizations to try to answer the proposed questions. If you realize some questions cannot be easily answered, this is totally fine. Just make a note in the process book and explain why.
2. Write down a list of important or interesting findings you found from the visualizations. The more, the better, but at least 3-4 per team member. Those will be important resources for you to build up your data story later. Record these findings as well as screenshots of your visualizations in your process book with each team member's name.
3. In the process, you may want to ask questions that are not in your original list, and this is entirely fine, keep a record of them in your process book.

Please document the above process in your process book.

Storyboard (Week 4)

1. Pick your main message.

Main message: Students with past class failures are at much greater risk of low final grades, so the school board should support early identification, targeted assistance, and evaluation of whether that assistance improves outcomes.

Reasons we chose this message: Past failures produced the clearest actionable separation in the explored data: 49.0% of students with any prior failure scored below 10, compared with 9.3% of students with none. Findings about higher-education plans, home internet, and guardian status add context, while family size, nursery attendance, and paid classes were weak or flat. This focus gives the school board a concrete signal for early outreach and an outcome it can track when evaluating support.

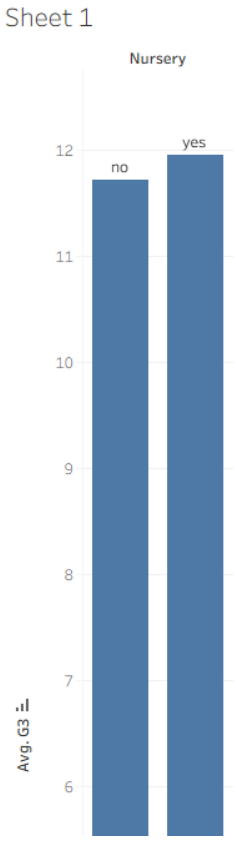
1. As a team, look at and discuss the list of insights you came up with. Pick one main insight that you think is most important or interesting to your audience. Formulate the insight as a message ("so what") in one sentence. You can choose multiple insights as long as they are coherent.
2. Record your main message and the reasons that your group chose it in your process book.
3. Sketch your data storyboard. Following the four steps that you learned in the lecture. You will create a data storyboard to communicate your main message. You can use pen&paper, which is the preferred option. You can also use Miro with the following steps.
4. You will work with your team to create the story arc. Please make sure that your story arc is lined up with the four elements of a data story (hook, rising insights, main message, and solution), similar to what we showed in the lecture.
5. It is fine if you have multiple insights for each data story element, again as long as they are coherent.
6. Take a screenshot of the final data arc (or a photograph of your papers) and put it in your process book.

Please document the above process in your process book.

Josh Blackmon Visualizations:

Q: Does having gone to nursery school matter in terms of grades?

Finding: There is a negligible difference between final grades between students who went to nursery school and those that did not, as seen in the bar chart below graphing G3 by nursery school attendance.



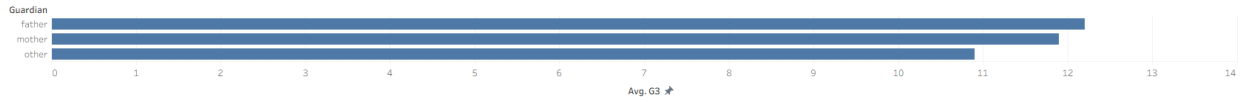
Q: Which of family support for education and school support has a greater impact on student grades?

Finding: Family support has a slightly positive correlation with grades, while school support has a negative correlation with grades. Students who had both seemed to be similar in grades to those that had neither. Figure below shows G3 scores on average for those that have Family support and/or school support.

	Schoolsup	
Famsup	no	yes
no	11.737	10.789
yes	12.140	11.469

Q: Does the guardian status matter for grades?

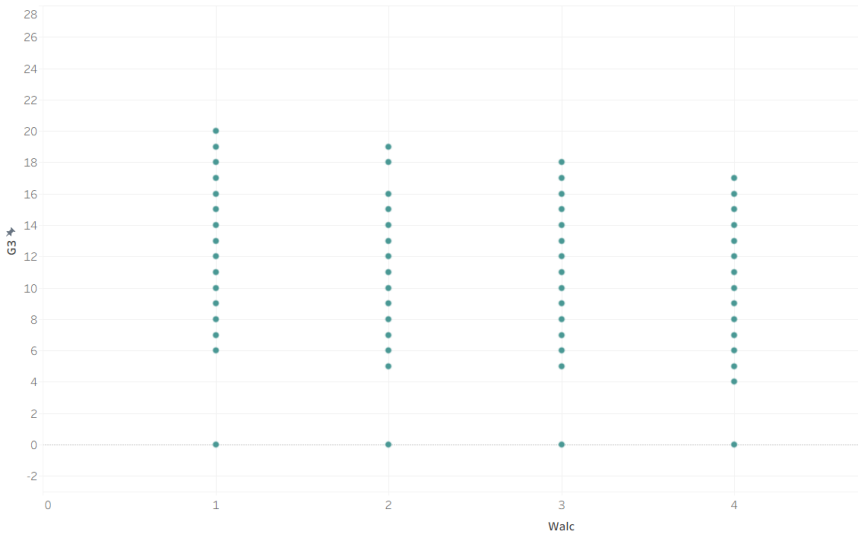
Finding: Students who have a father or mother for a guardian had very similar G3 scores, while those that had a non parent guardian had a noticeable drop in G3 score. Figure below shows G3 scores on average grouped by guardian type.



Yash Agrawal Visualizations

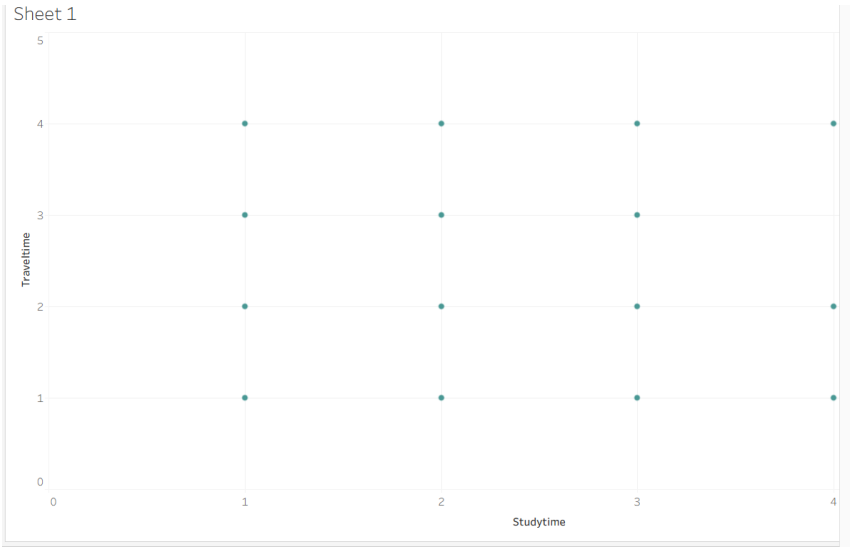
Q: Is there a relationship between alcohol consumption and grades for these students?

Finding: Below is weekly alcohol usage vs G3 grades, as noted there's a slight downwards trend suggesting the more alcohol usage results in lower grades.



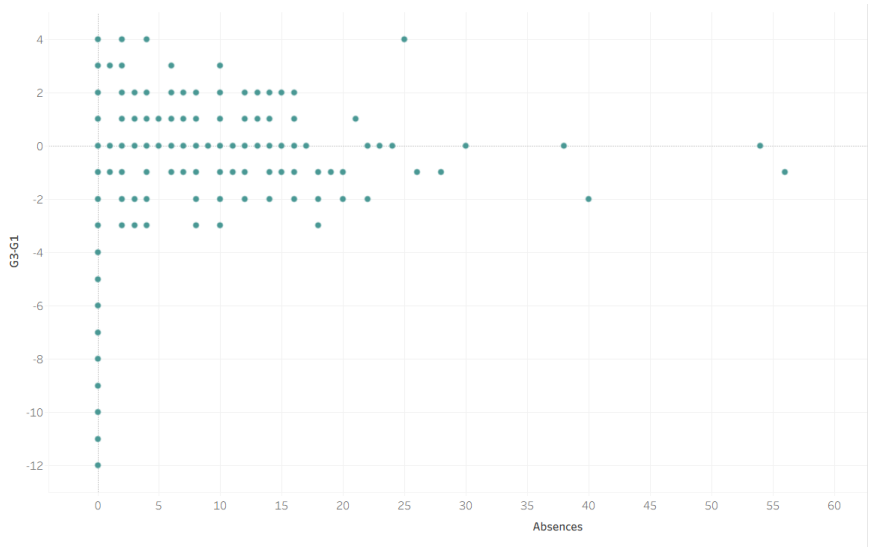
Q: Is there an inverse relationship between study time and travel time?

Finding: Below is the study time vs travel time graph and there seems to not be much correlation at all, suggesting that there is no correlation between the two, which is interesting because you would think travel time would take away from study time.



Q: Do absences correlate with lower grades throughout the year?

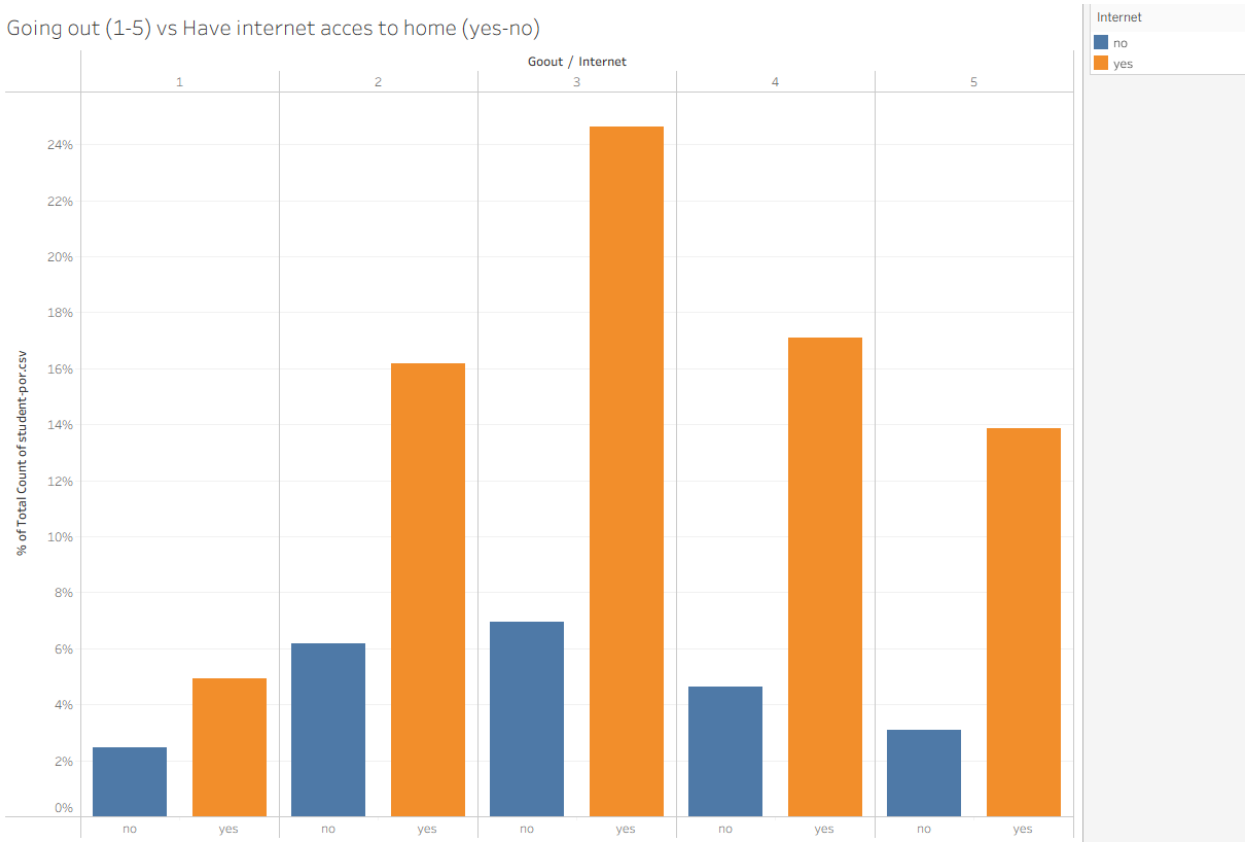
Finding: Below is the absences vs the change in grade from G1 to G3. Theres a slight negative correlation where more absences reduce grade.



Justin Bimal Visualizations:

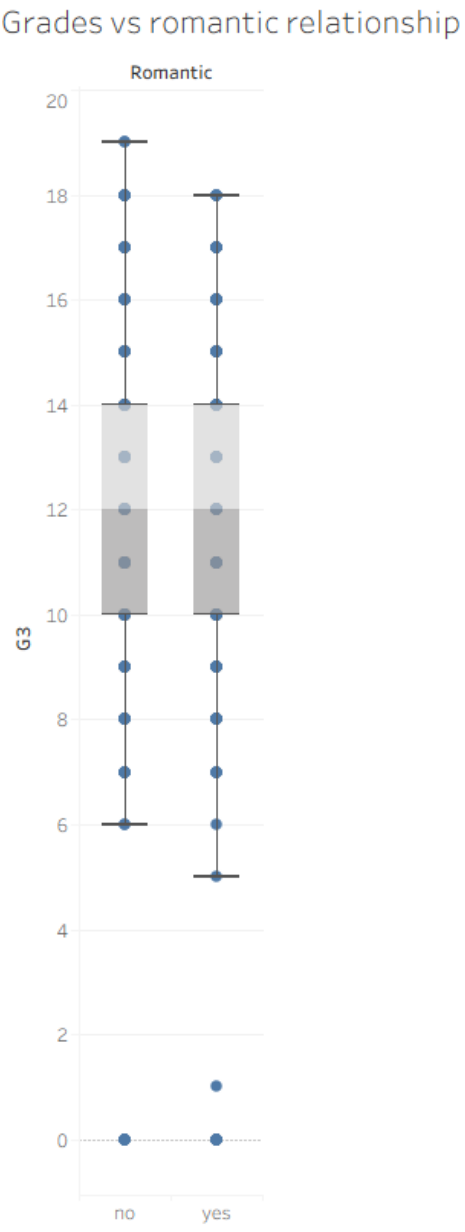
Q: Are internet access and the scale of going out related? Do students without internet access go out more?

Findings: Students with internet access seem to go out at similar patterns rather than more frequently, though they make up the large majority of the total population. Students are most concentrated at going-out level 3 no matter if they have internet access home or not. Level 1 of going out is the smallest percentage for both groups. Students without internet access do not appear to go out more than those with internet access.



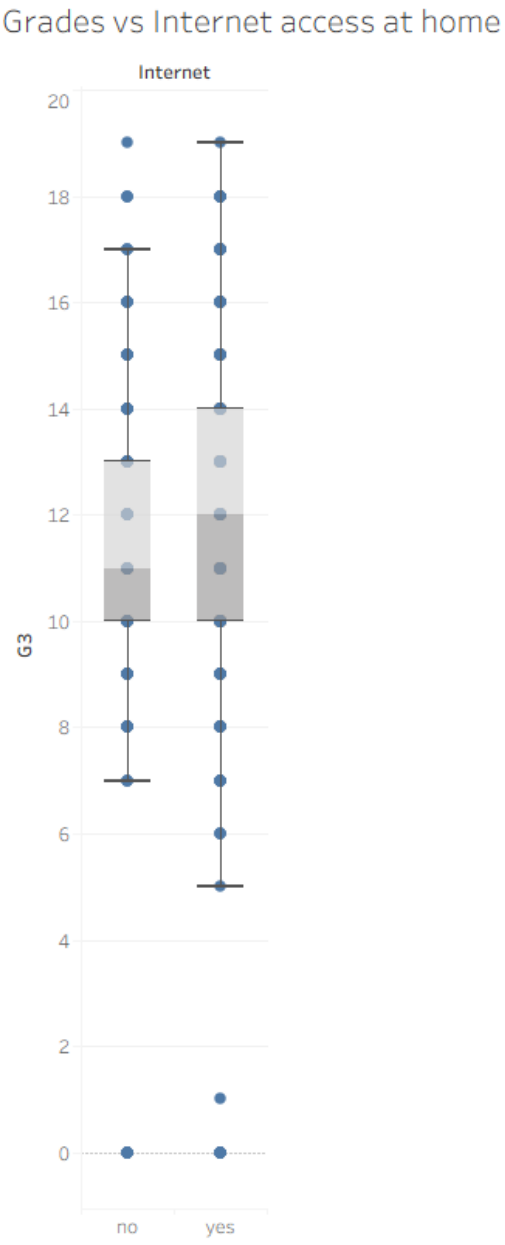
Q: Does relationship status have a correlation with grades?

Findings: Students who are not in a romantic relationship have the same median grade: around 12. The maximum grades reached by single students are slightly higher, reaching up to 19, while students in a relationship reach around 18. Both groups share the same upper quartile of 14 and lower quartile of 10. Romantic relationship shows almost no association with grades.



Q: Does student internet access at home have a correlation with grades?

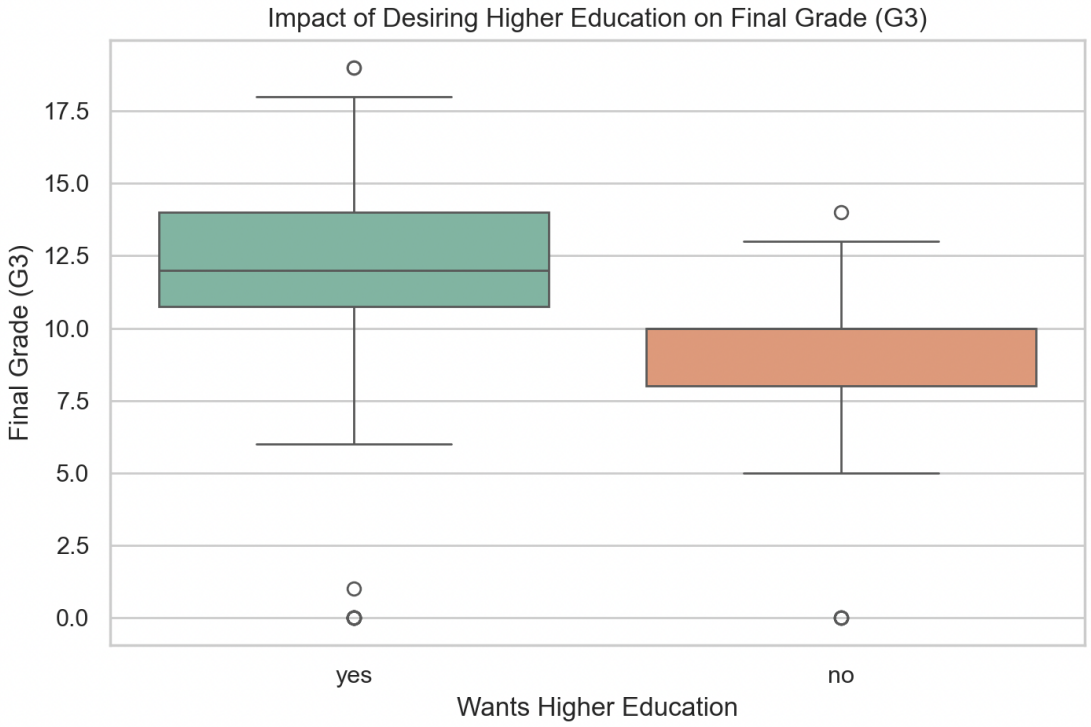
Findings: Students with home internet access get a higher median grade than those without internet access, around 1 point of difference. The upper whisker for students with internet access reaches higher at around 19 compared to around 17 for those without. The upper quartile is also 1 point higher for students with internet, while both groups share the same lower quartile of 10. This suggests that lacking home internet may be a disadvantage for reaching higher grade. Home internet access has a modest positive impact on grade.



Timi Omoteso Visualizations:

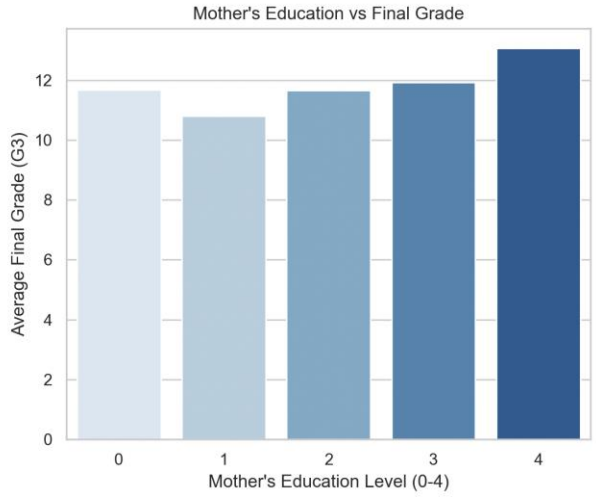
Q: Does wanting to pursue higher education result in higher grades?

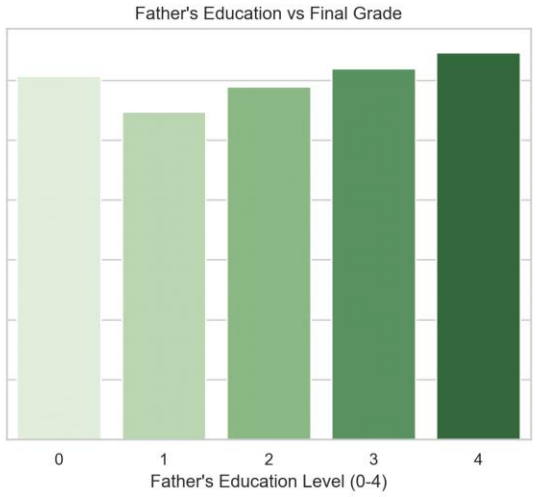
Yes, it makes a significant difference. Students aiming for higher education have a median grade of 12.0, with their middle 50% of scores falling between roughly 10.5 and 14.0. Those who are not planning on higher education have a lower median of just 8.0, with the bulk of their grades clustering tightly between 8.0 and 10.0.



Q: Whose education status matters more for the purposes of grades: Mom or Dad?

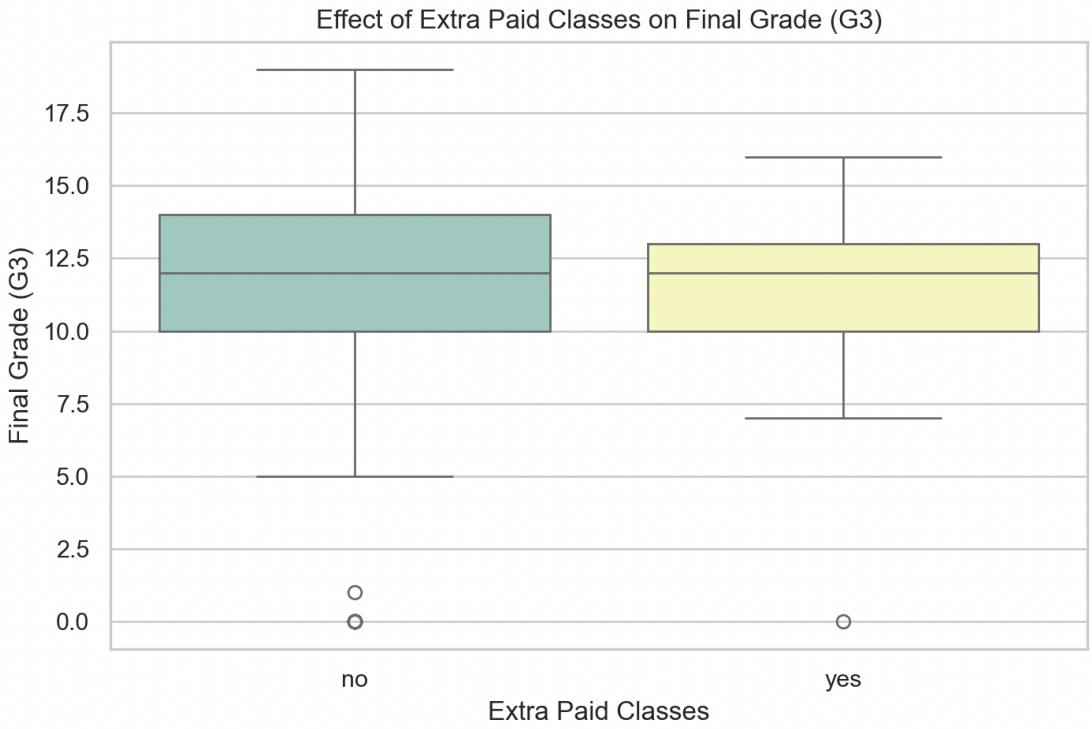
Findings: Both parents' education levels follow a nearly identical trend, meaning neither has a distinctly stronger impact than the other. For both the mother and the father, the student's average final grade starts around 11.8 to 12.2 at education level 0, dips to roughly 10.8 to 11.0 at level 1, and then steadily climbs to peak at 13.0 at education level 4.





Q: Which controllable factor (like extra paid classes) leads to the most increase in grades?

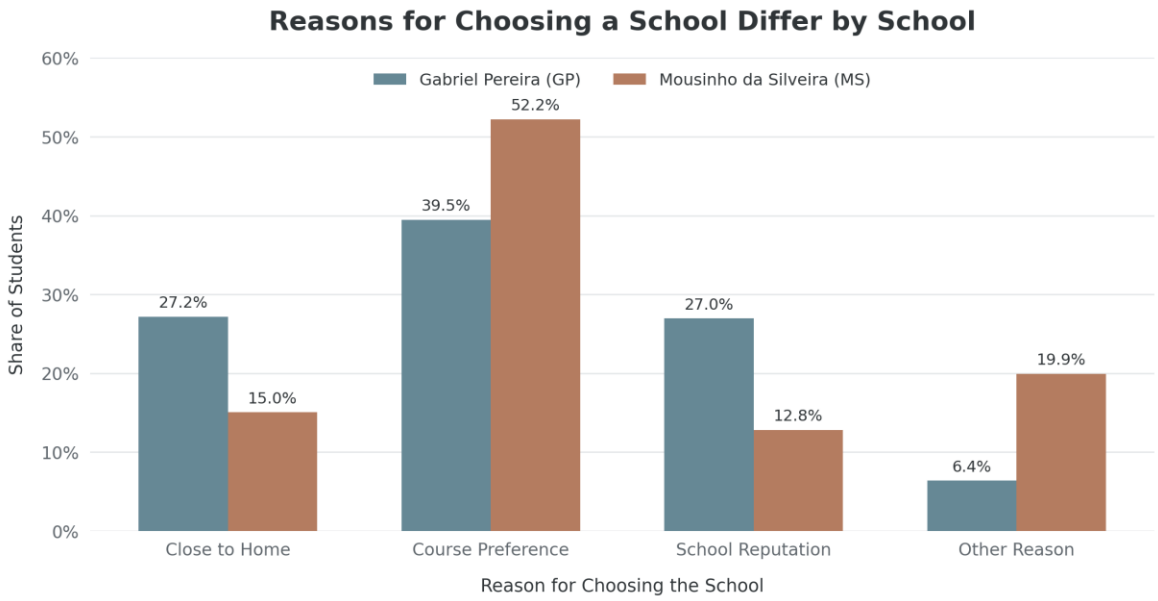
Findings: Surprisingly, extra paid classes do not give a huge edge. The median final grade is exactly 12.0 whether a student takes paid classes or not. Furthermore, the group without paid classes actually has a slightly higher upper quartile (14.0) compared to the paid group (13.0), showing that paying for tutoring does not automatically yield better grades.

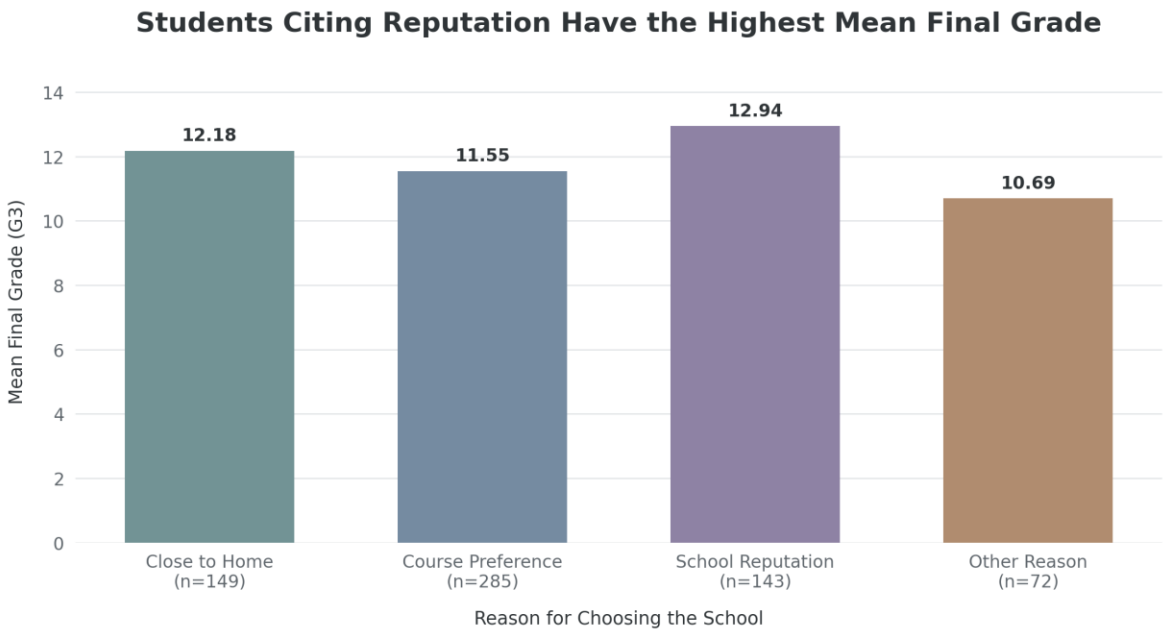


Abdulaziz Alajlan Visualizations:

Q: What school has a higher “home” preference rate? How about for “course” or “reputation”? Is that relevant for grades?

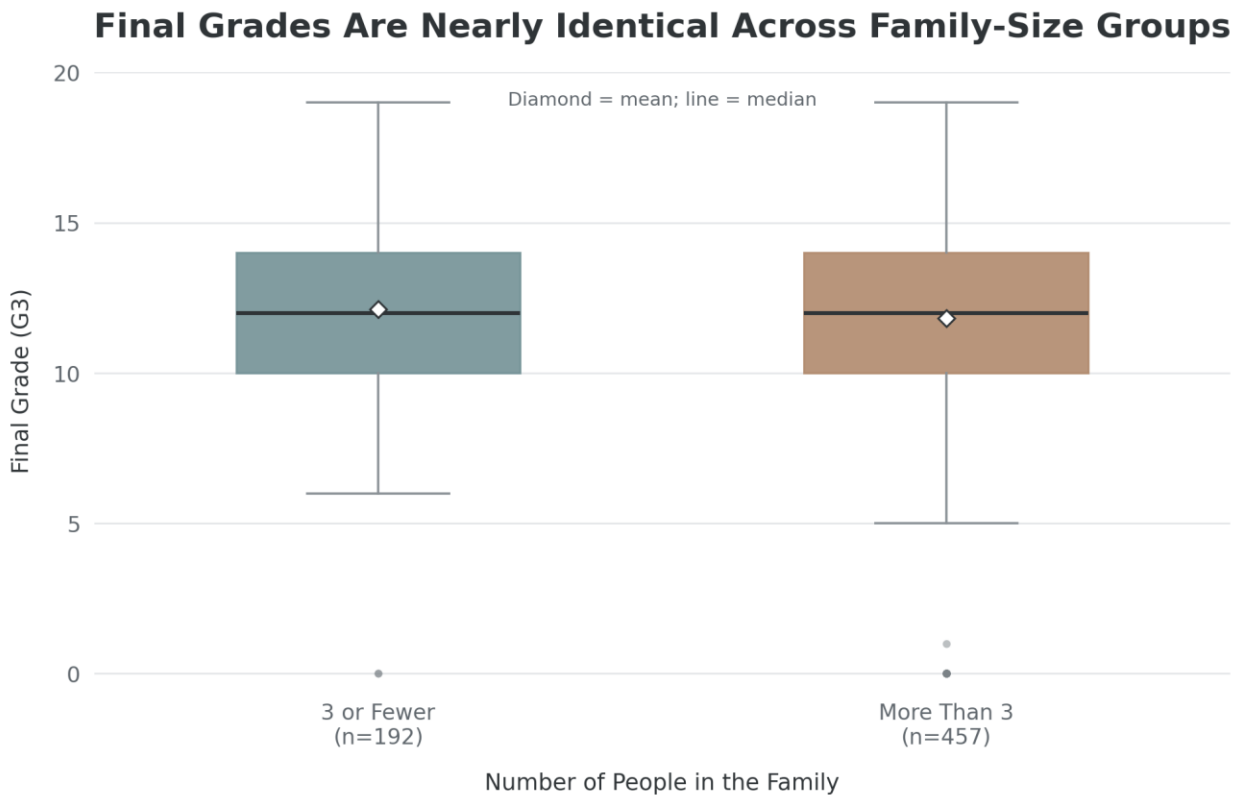
Finding: Gabriel Pereira has higher home (27.2% versus 15.0%) and reputation (27.0% versus 12.8%) preference rates, while Mousinho da Silveira has the higher course-preference rate (52.2% versus 39.5%). Across both schools, students who chose for reputation have a 93.0% pass rate (Pass rate means the percentage of students with G3 ≥ 10; below the pass mark means G3 < 10), compared with 88.6% for home, 80.7% for course, and 75.0% for other reasons; reputation and other also have the highest and lowest mean final grades (12.94 and 10.69). Because both the reason mix and grades differ by school, administrators should treat reason as a follow-up screening clue—not evidence that the stated reason causes the grade difference.





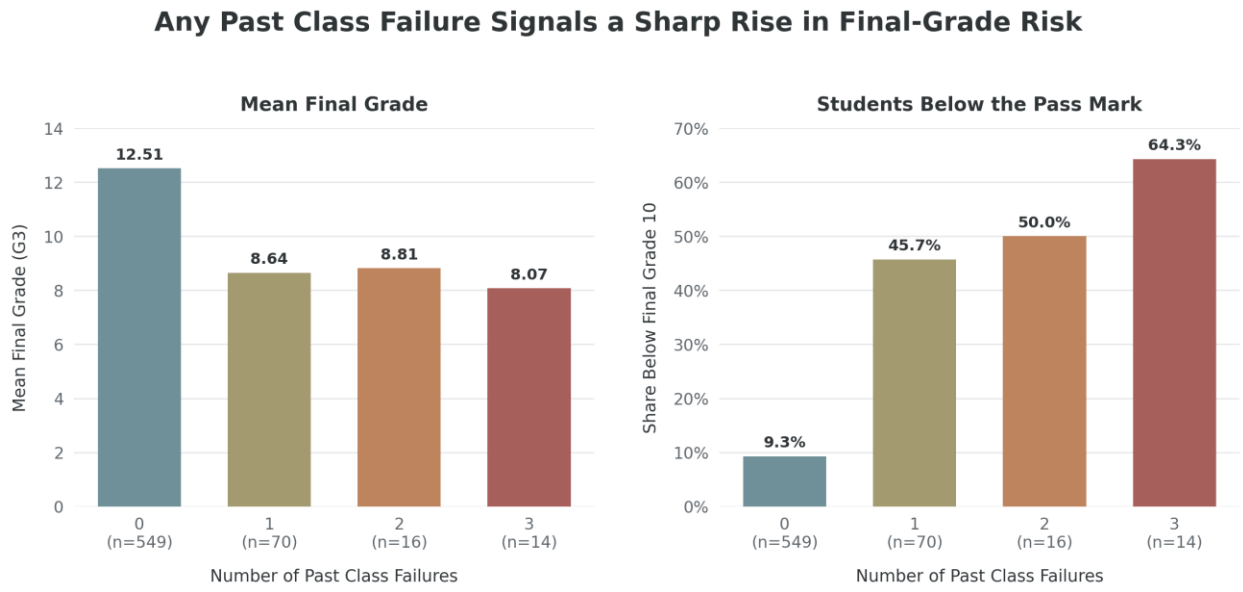
Q: Does familial size correlate to grades?

Finding: The grade distributions are essentially flat across family size: both groups have a median of 12 and an interquartile range of 10–14. Students from families of three or fewer pass at 87.5%, compared with 83.4% for larger families—a modest 4.1-percentage-point difference—and their mean grades differ by only 0.32 points. Family size alone is therefore not a strong basis for administrators to target academic support.

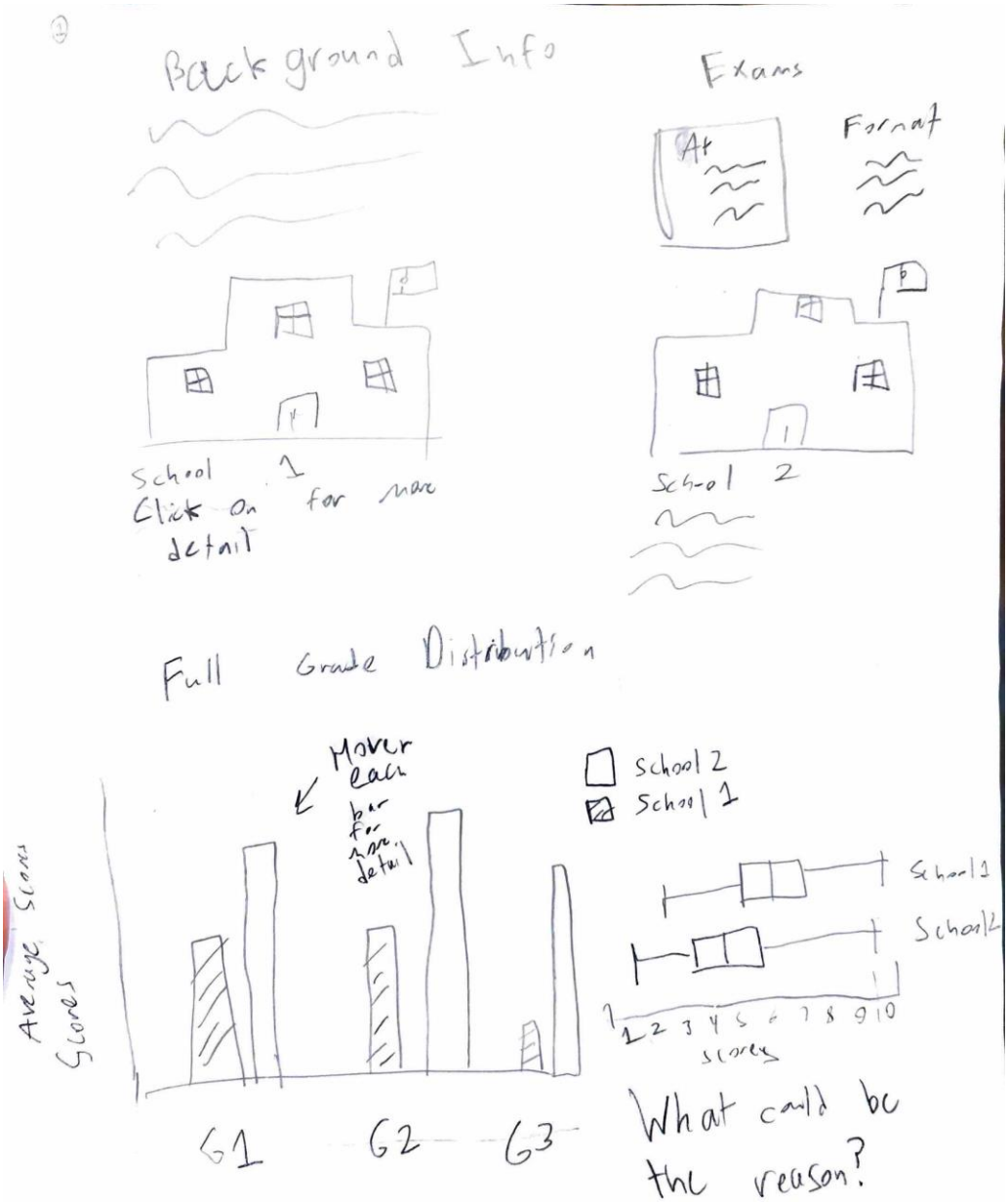

Final Grades Are Nearly Identical Across Family-Size Groups
Diamond = mean; line = median
Final Grade (G3)
20
15
10
5
0
3 or Fewer
(n=192)
More Than 3
(n=457)
Number of People in the Family

Q: Do past class failures predict final grades?

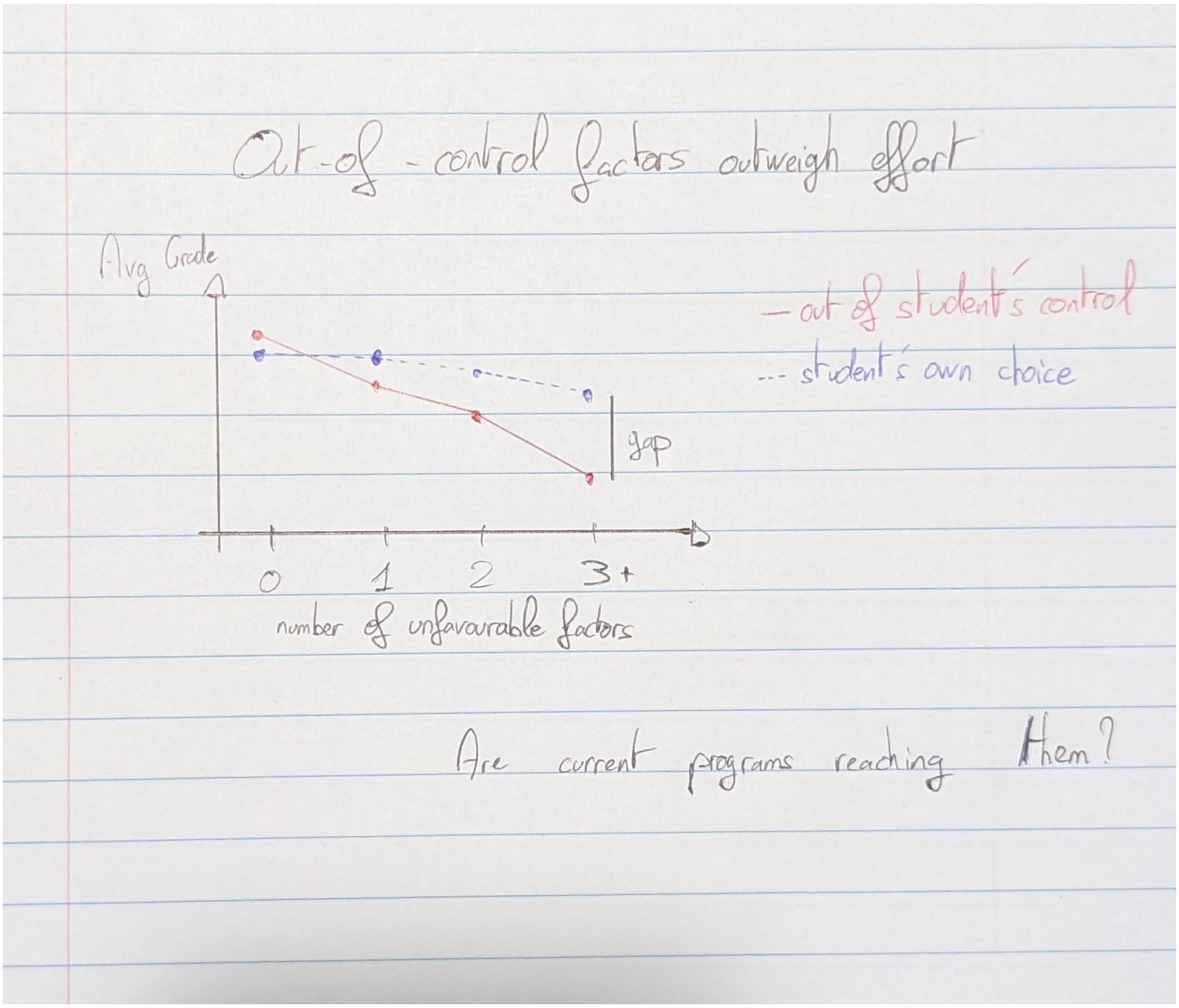
Finding: The share below the pass mark rises from 9.3% for students with no past failures to 45.7%, 50.0%, and 64.3% at failure levels one, two, and three; mean final grades are 12.51, 8.64, 8.81, and 8.07, respectively. Combining the nonzero levels, 49 of 100 students with any past failure fall below 10, compared with 51 of 549 students with none—a 39.7-percentage-point risk difference. Administrators could use any recorded failure as an early-support flag, while avoiding fine distinctions between levels two and three because those groups contain only 16 and 14 students.

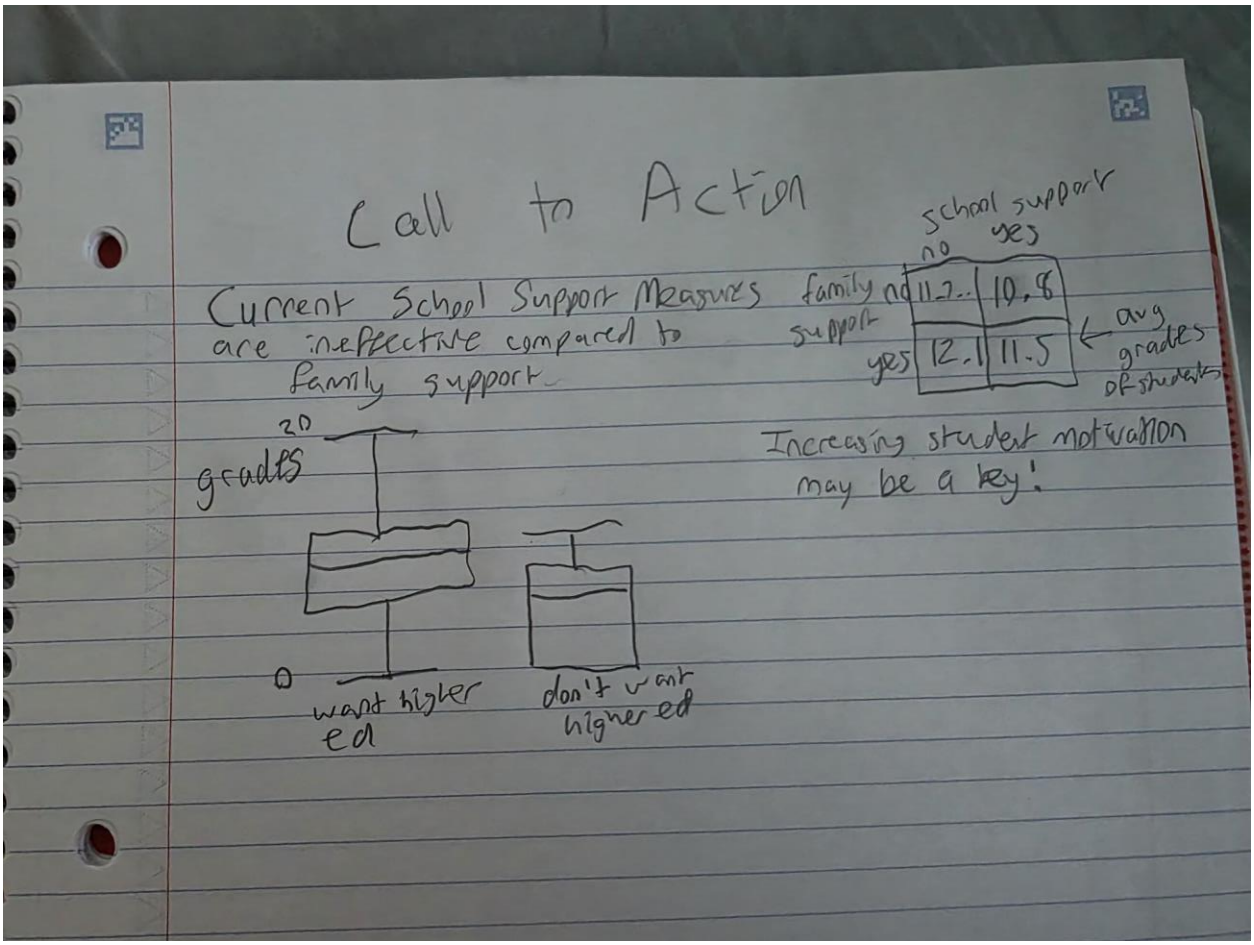


Background:


Background Info
Exams
At
Format
School 1
Click on for more detail
School 2
Full Grade Distribution
Hover each bar for more detail
School 2
School 1
Average Scores
G1
G2
G3
School 1
School 2
1 2 3 4 5 6 7 8 9 10
scores
What could be the reason?

Call to Action:

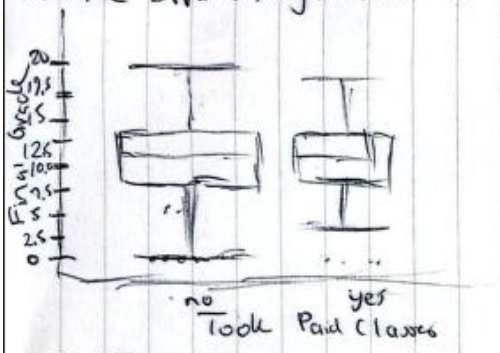




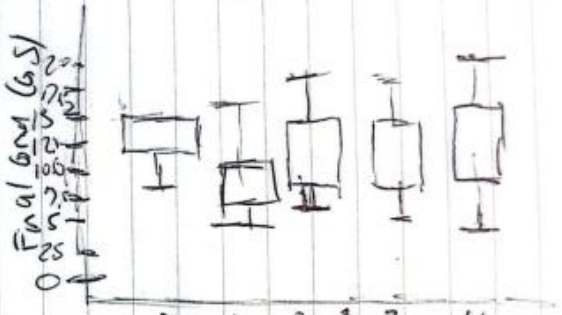
Storyboard (Week 4) — Completed Story Arc

Rising Action

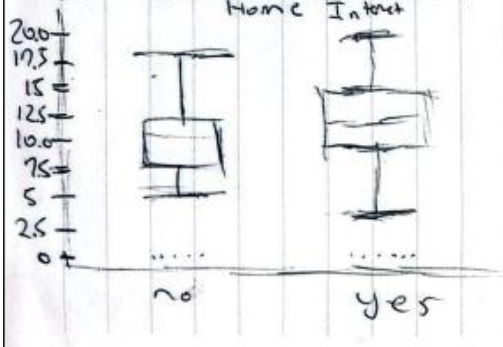
1. The 'Effort' Myth: Extra Paid Classes



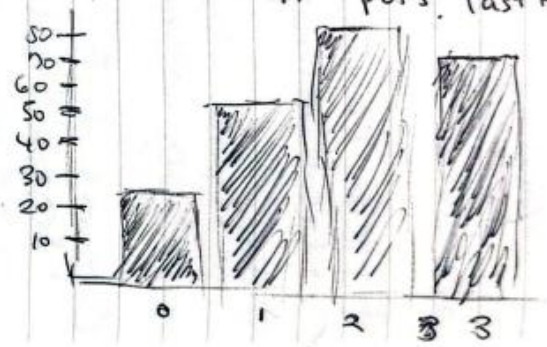
Systemic Advantage: Motor Educ



3. The Infrastructure Divide: Home Internet



4. Admin Blind Spots: Past F...



Hook: Nursery attendance has a negligible relationship with grades (Joshua), family-size groups have identical medians and middle-50% ranges (Abdulaziz), and paid classes show no meaningful grade advantage (Timi). These non-results challenge the assumption that familiar demographic or financial factors automatically identify the students at most risk.

Rising insights: Students planning to pursue higher education have higher grades (Timi), students with home internet have a roughly one-point median advantage (Justin), and students with a non-parent guardian show a noticeable grade drop (Joshua). These findings narrow the story from broad assumptions to indicators connected with student goals, access, and home context.

Main message: Among students with any past failure, 49.0% score below 10 and mean G3 is 8.59; among students with no past failure, 9.3% score below 10 and mean G3 is 12.51. Past academic history therefore provides the school board with a concrete group for early attention.

Solution (recommendation, not a finding): Use prior-failure records to begin early outreach, assess whether internet access or guardian circumstances create additional needs, offer targeted academic support, and track later outcomes to evaluate whether the support works. The lower observed average among school-support recipients may reflect that support is assigned to students already struggling, not that support harms grades.

Sketches

Message ID Chart

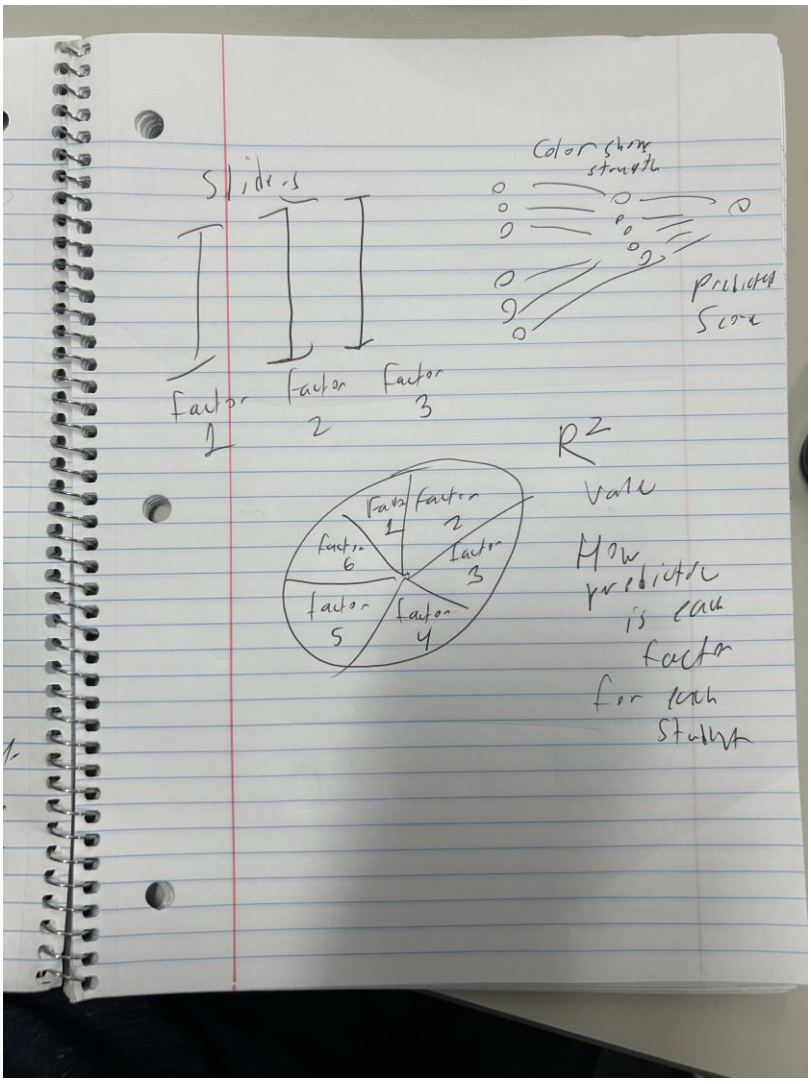
ID Meaning	Message
1	Describe how it's very unclear to be able to predict student grades. Showcase that there is a variety of factors going on and a holistic view that deeper analysis is needed. Used for the Hook
2	Describes the whole distribution of students and show the high variation of grades between schools. Intrigues reader of why is there is such a difference. Used for hook
3	Showcase each student’s journey throughout the year as how their grades changed.
4	Showcase that study hours did not have much effect as fully imagined for student;s grades
5	Structural factors (home internet access, daily commute, guardian status) correlate

	directly with an immediate and lasting grade gap.
6	When evaluated on identical scales, only specific external barriers strongly correlate with grades, while several assumed drivers show negligible impact.
7	Out-of-control systemic constraints outweigh personal effort: even with equal effort, accumulating disadvantage strongly depresses student outcomes
8	School support programs show a critical mismatch: the students with the highest disadvantage level do not receive support proportionally.
9	Students who desire higher ed have significantly higher grades
10	Nursery attendance, Gender, Family Size, Money spent on outside classes has no correlation with grades
11	Students with even a single past failure have significantly lower grades, decreasing further per failure
12	Grades fall as absences rise: students with 16 or more absences average about 1.4 points below students with 7 or fewer.
13	The two schools differ in why students chose them (Gabriel Pereira: home and reputation; Mousinho da Silveira: course), and reputation-choosers pass at 93% versus 75% for "other." Reason is a screening clue, not a cause.

Sketch ID	Message ID	Author	Vote
1	1	YA	
2	1	YA	
3	2	YA	
4	2	YA	1
5	3	YA	
6	4	YA	
7	1	YA	11
8	3	YA	

9	2	JBL	1
10	5	JBL	111
11	6	JBL	
12	5	JBL	1
13	4,7	JBL	
14	8	JBL	1 1
15	6	JBL	
16	7	JBL	111
17	10	AJ	1
18	10	AJ	
19	9	AJ	
20	12	AJ	
21	4	AJ	
22	5, 9	AJ	1|1
23	11	AJ	
24	13	AJ	
25	9	JB	
26	10	JB	1
27	10	JB	
28	5	JB	1
29	11	JB	1 1
30	8	JB	1
31	10	JB	
32	5	JB	
33	10, 6	TO	
34	5	TO	
35	9	TO	
36	5	TO	
37	11	TO	1
38	11	TO	
39	10	TO	
40	8,5, 11	TO	111

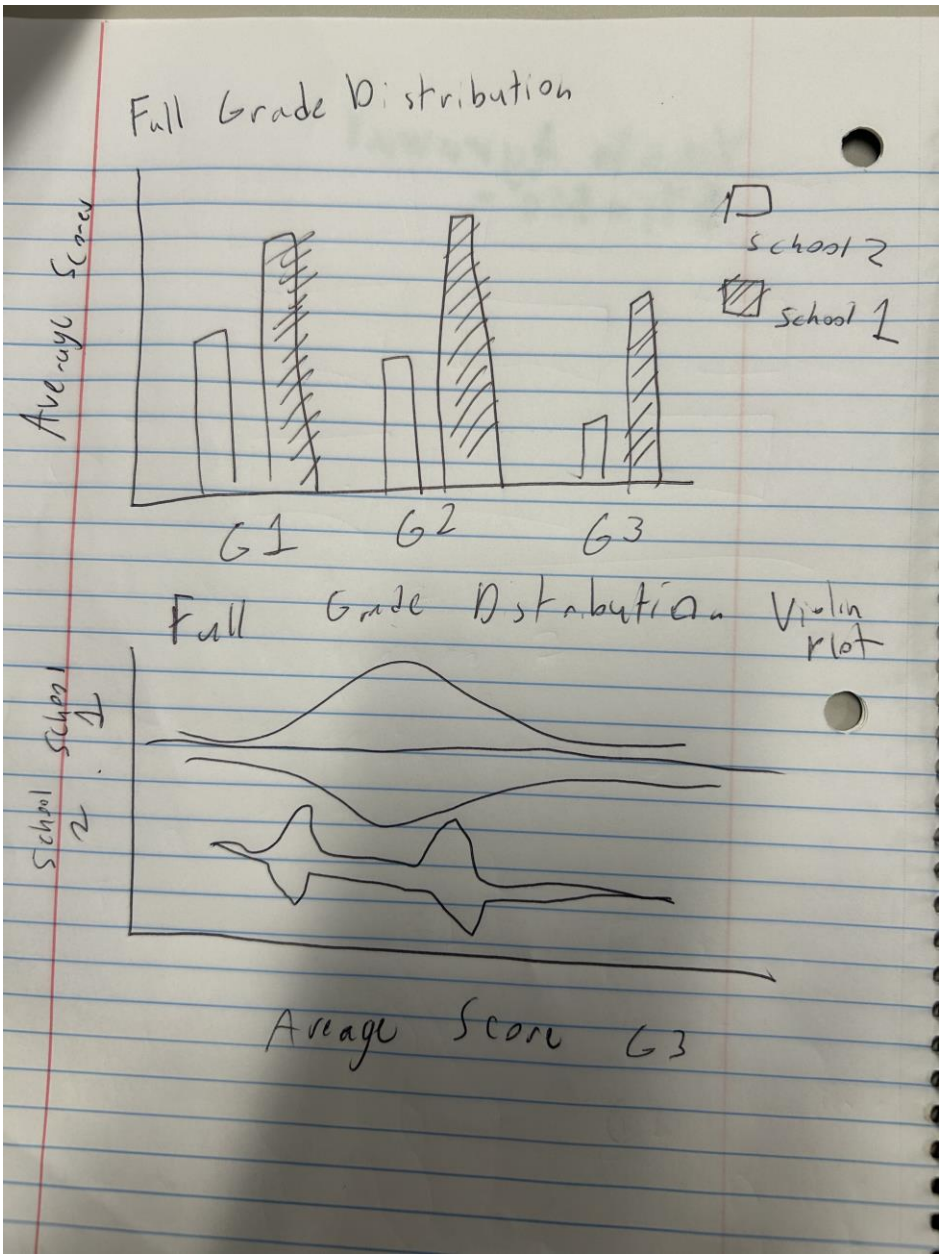
Yash Agrawal Sketches



Sketch ID 1 – Yash Agrawal – Message ID 1. This graphic idea is that we train a simple machine learning model, and we have a bunch of slider factors that a user can try to move around and see how that effects the predicted score based on the data. We would show the regression model on the right showing which neurons are activated and which weights matter most based on color and the idea is that the user is intrigued as they play with it that factors like study hours do not matter as much they would imagine. Would be used for the hook.

Sketch ID 2 – Yash Agrawal – Message ID 1. This graphic would be a pie chart graphing the R^2 value for each of the factors to see how well it predicts or explains grades throughout the year, and we would have one of the parts of the pie chart to be just unknown. The idea

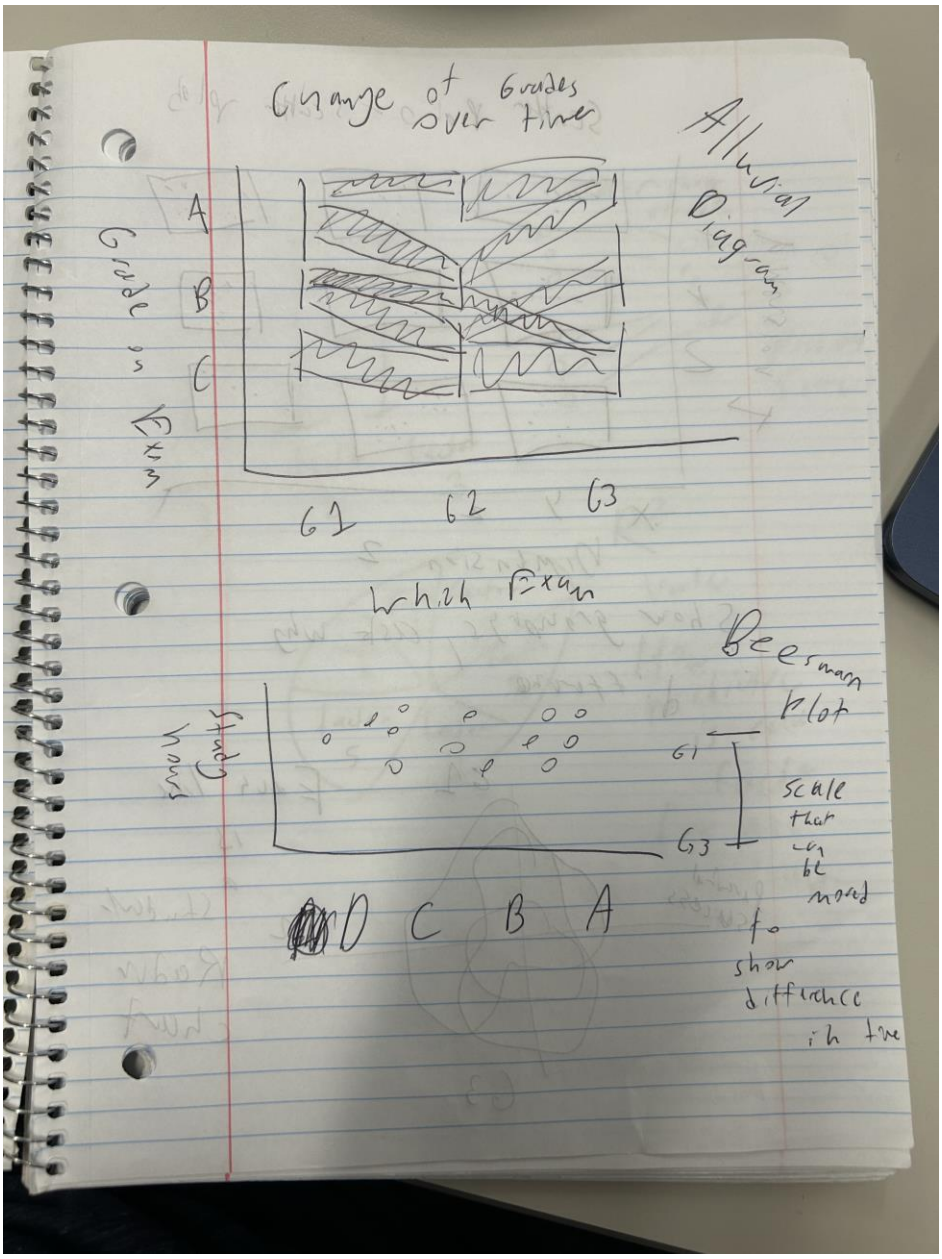
is that it would be interesting to analyze further of why do some factors have very little explainability and wonder what further analysis we can do. Would be used for the hook.



Sketch ID 3 – Yash Agrawal – Message ID 2. This diagram is a simple bar chart comparison of the schools and their average grades between G1, G2, and G3. The message is that it will show that there is a significant difference in the grades between the schools despite being part of the same district and of very similar populations, and so it will make the reader wonder what is going on for purposes of the hook. Would be used for the hook.

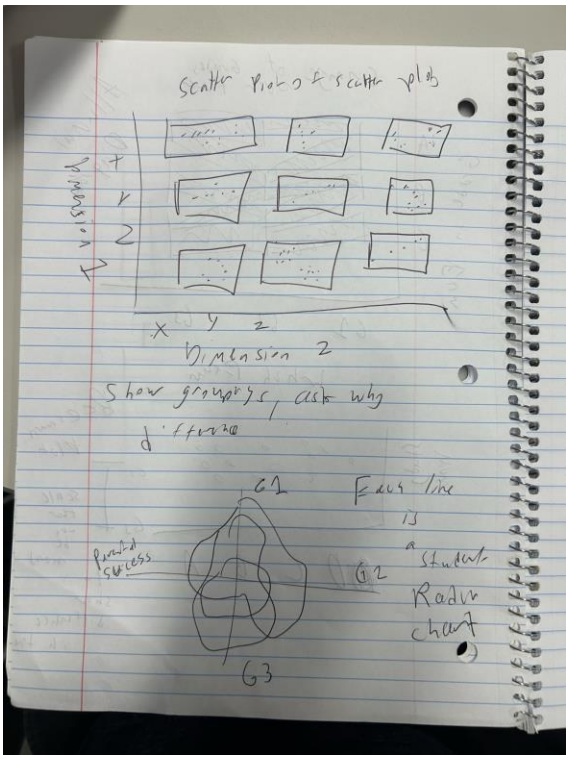
Sketch ID 4 – Yash Agrawal – Message ID 2. This diagram would be a violin plot of the distribution of grades per school, and possibly we would label big centers of the

distribution of what those students represent. It would help again be part of the hook for students to wonder what further analysis can be done and why the distributions are like this. Would be used for the hook.



Sketch ID 5 – Yash Agrawal – Message ID 3. This diagram would be an alluvial plot showcasing each student's journey from their grades from G1 to G3. So a student might go from A to B to C over the year and this plot would show that there is some questions about why there is so much change in grades throughout the year. That would imagine that some consistency for high performing students or not. This gives an overall view of the grade distribution and raises questions about why this is. Would be used for the hook.

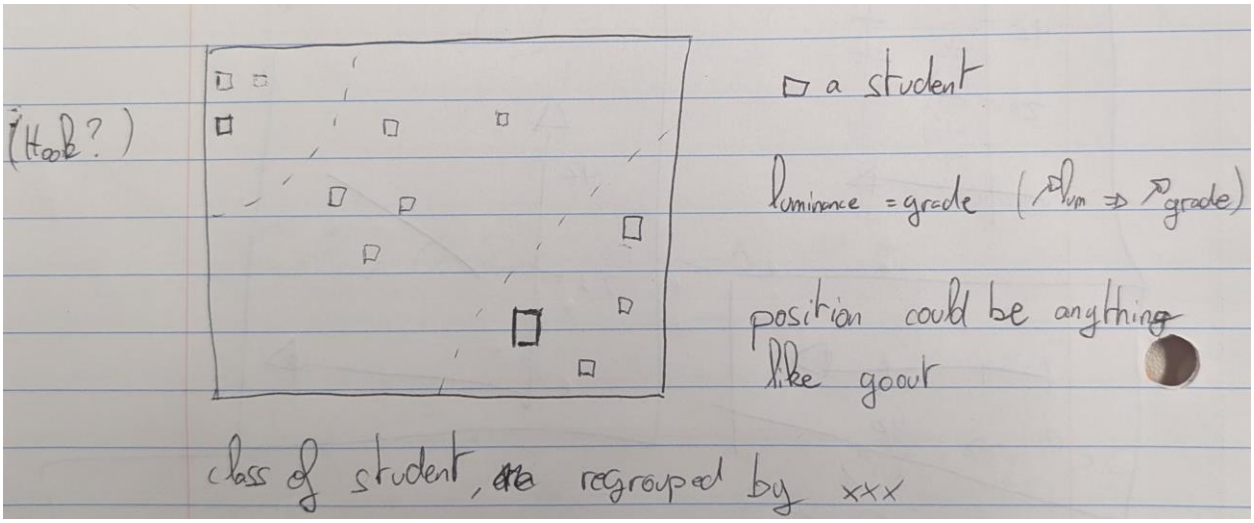
Sketch ID 6 – Yash Agrawal – Message ID 4. This diagram is a beeswarm plot where each dot represents a student and the user can move the slider from G1 to G3 to see how students move. We can color students by some external factor such as parental occupation and the y column is study hours and we can see how each student moves throughout the school year. Would be used for the hook to describe overall student distribution. Would be used for the rising action.



Sketch ID 7 – Yash Agrawal – Message ID 1. This diagram would show a scatterplot of scatterplots showing how each combination of factors are scatterplotted showing an overall uniqueness to how the factors may be correlated or not. We would highlight graphics that have grades associated with them, but the idea is that a reader would wonder why there are some groupings, and a student could click on each graph and see different interesting groups and plots. Would be used for the hook.

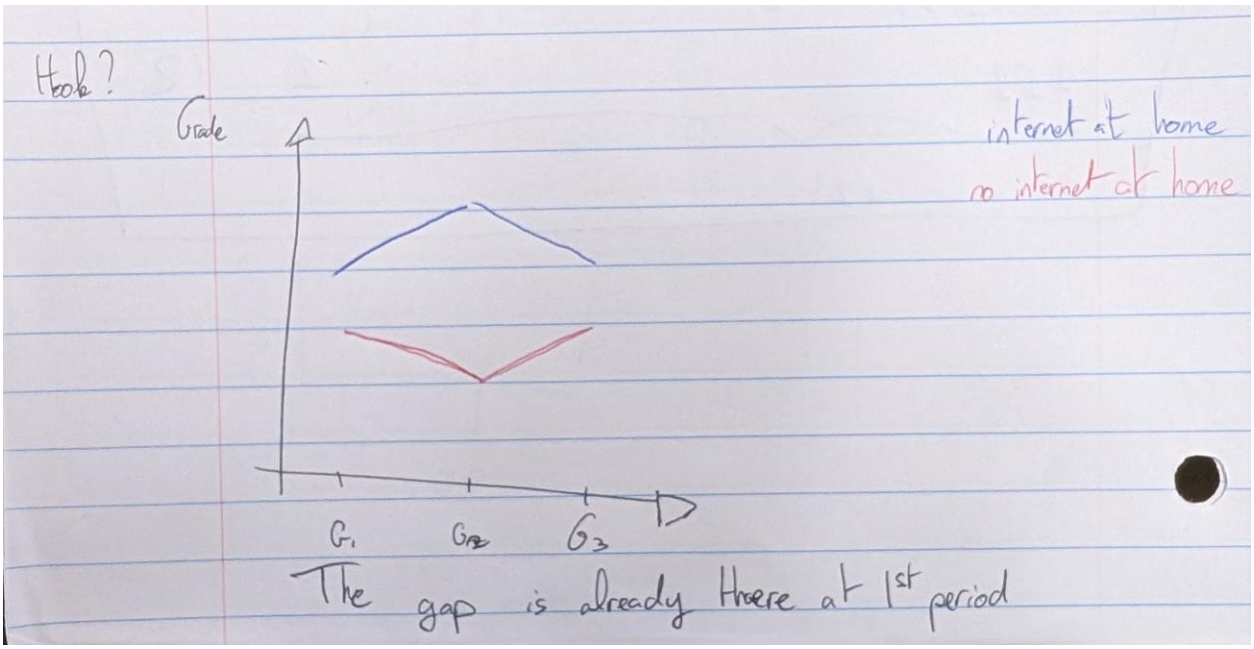
Sketch ID 8 – Yash agrawal – Message ID 3. This diagram would showcase each student's journey as a line around the graphic where it would be their grades throughout the semester and possibly some initial factor. The starting value might be parental success in some way and so we showcase the movement of each student throughout. Each student would be a different color so it would be interesting to see the chaos or order. Would be used for the hook.

- Sketch ID 9 – Justin Bimal– Message ID 2:



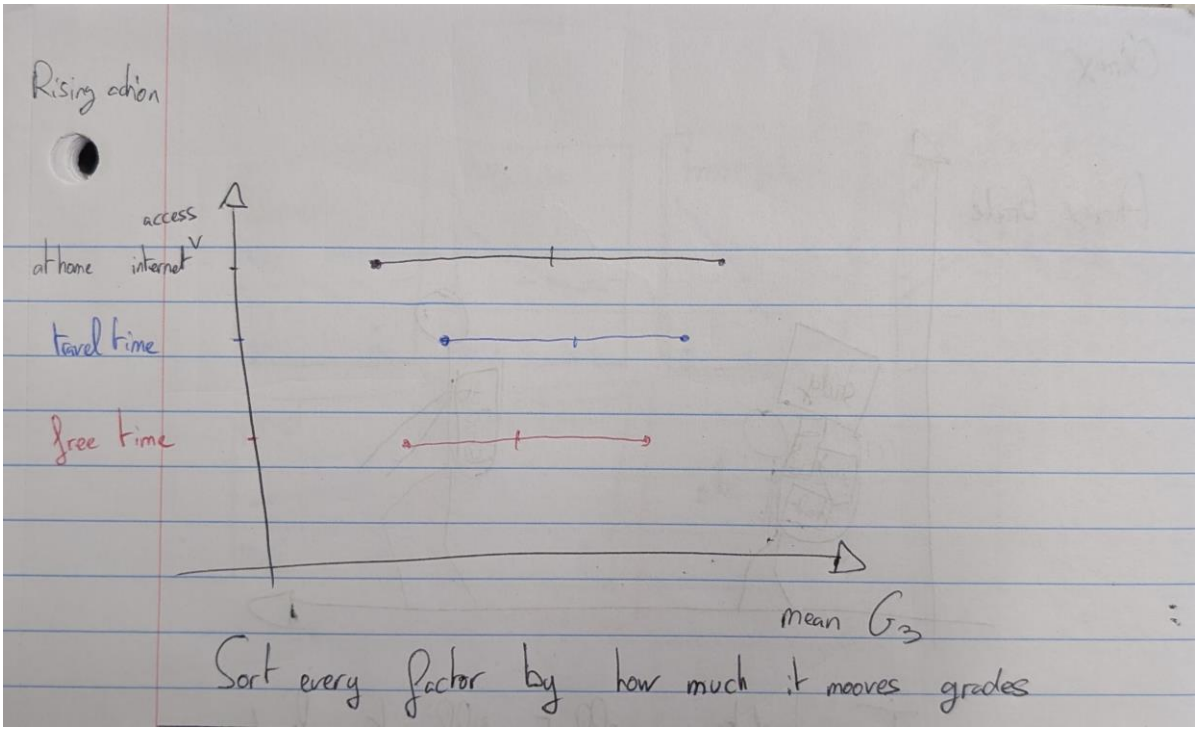
This sketch is intended for the hook. The main objective is to highlight the overall distribution and variations in scores within a cohort of students. By changing the spatial grouping, users can instantly see whether the lowest scores are concentrated under specific conditions.

- Sketch ID 10 – Justin Bimal– Message ID 5:



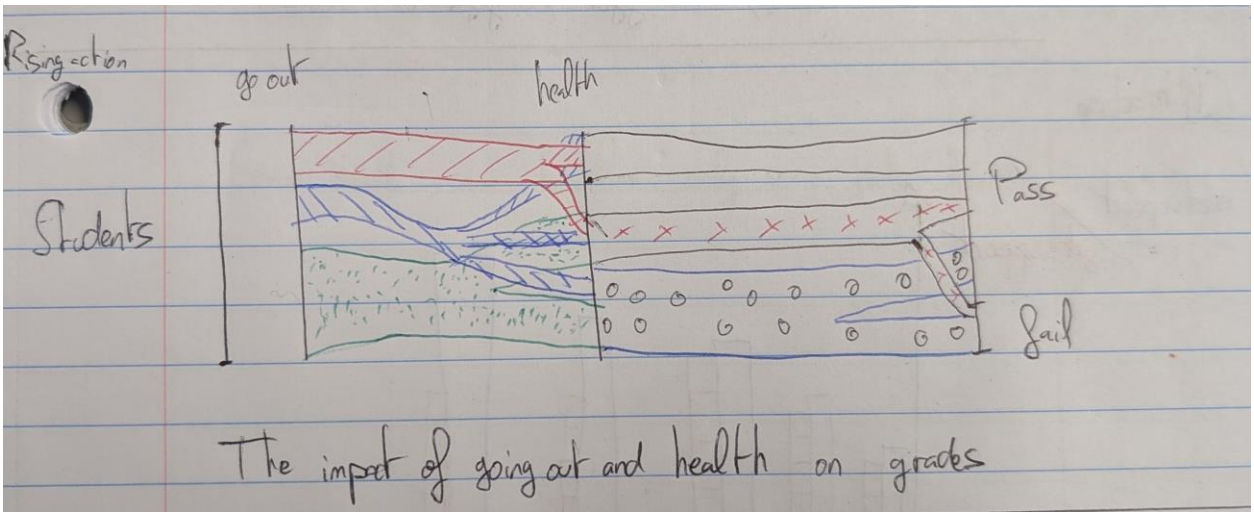
This sketch is intended for the hook. This chart uses line marks where horizontal position encodes the grades, vertical position encodes average grade, and color hue distinguishes students with and without home internet. The graphic highlights that the performance gap caused by digital infrastructure is already visible at the first grading period (G1) and remains persistent throughout the entire school year.

- Sketch ID 11 – Justin Bimal– Message ID 6:



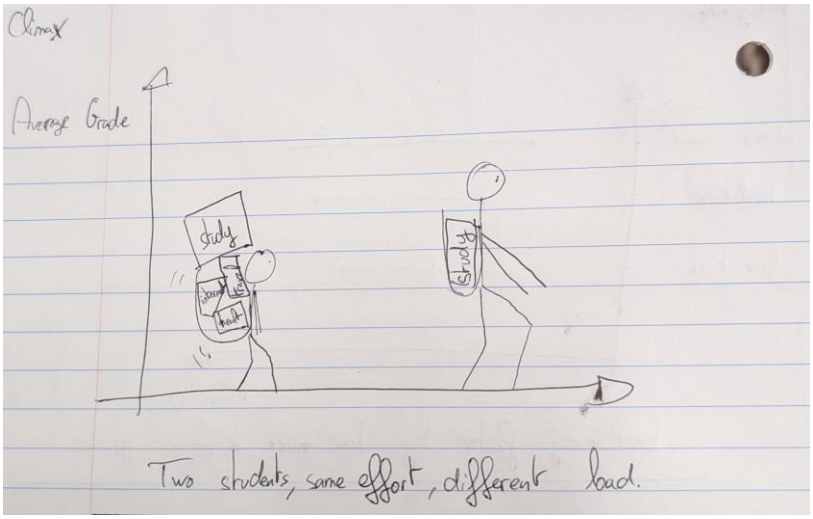
This sketch is intended for the rising action.This graph uses horizontal interval markers: the vertical position represents external factors, while the horizontal position/length corresponds to the average scores in G3 and the extent of variation. Factors such as home Internet access, commute time, and free time are ranked from highest to lowest based on their influence on final grades, allowing observers to quickly identify the variables that are actually correlated with student success.

- Sketch ID 12 – Justin Bimal– Message ID 5:



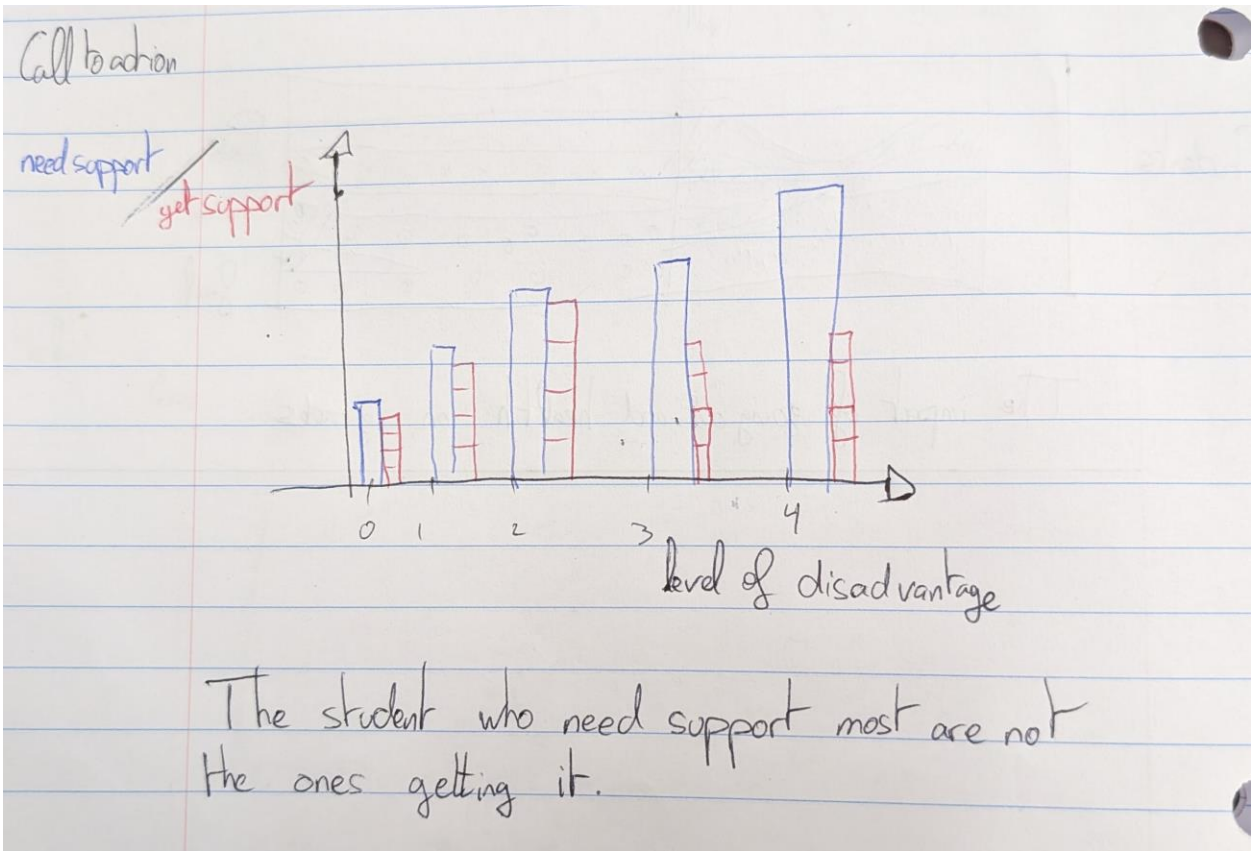
This sketch is intended for the rising action.This alluvial diagram uses connecting bands: the height and width of the stream represent the number of students, while the horizontal position categorizes successive groups: school absences, health status, and final academic outcomes (success or failure). It illustrates how combinations of social activities and health statuses branch out and correlate with rates of academic success or failure.

- Sketch ID 13 – Justin Bimal– Message ID 4, 7:



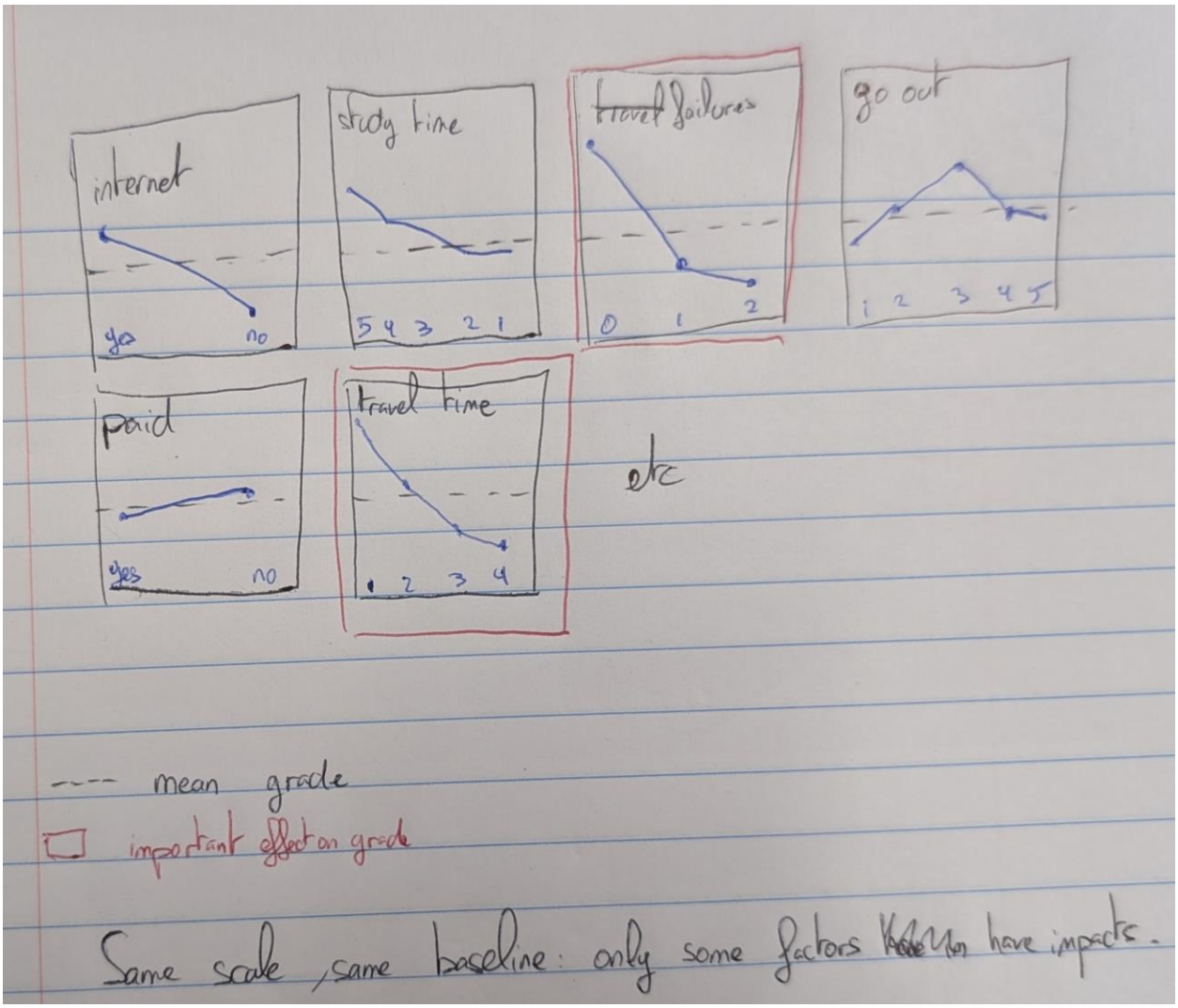
This sketch is intended for the main message. It uses figurative glyph marks where vertical position represents final grade and glyph size/load area encodes the weight of external disadvantages. It illustrates the central narrative contrast: two students dedicate the exact same study effort, but one carries extra structural burdens, explaining why equal effort does not produce equal grades.

- Sketch ID 14 – Justin Bimal– Message ID 8:



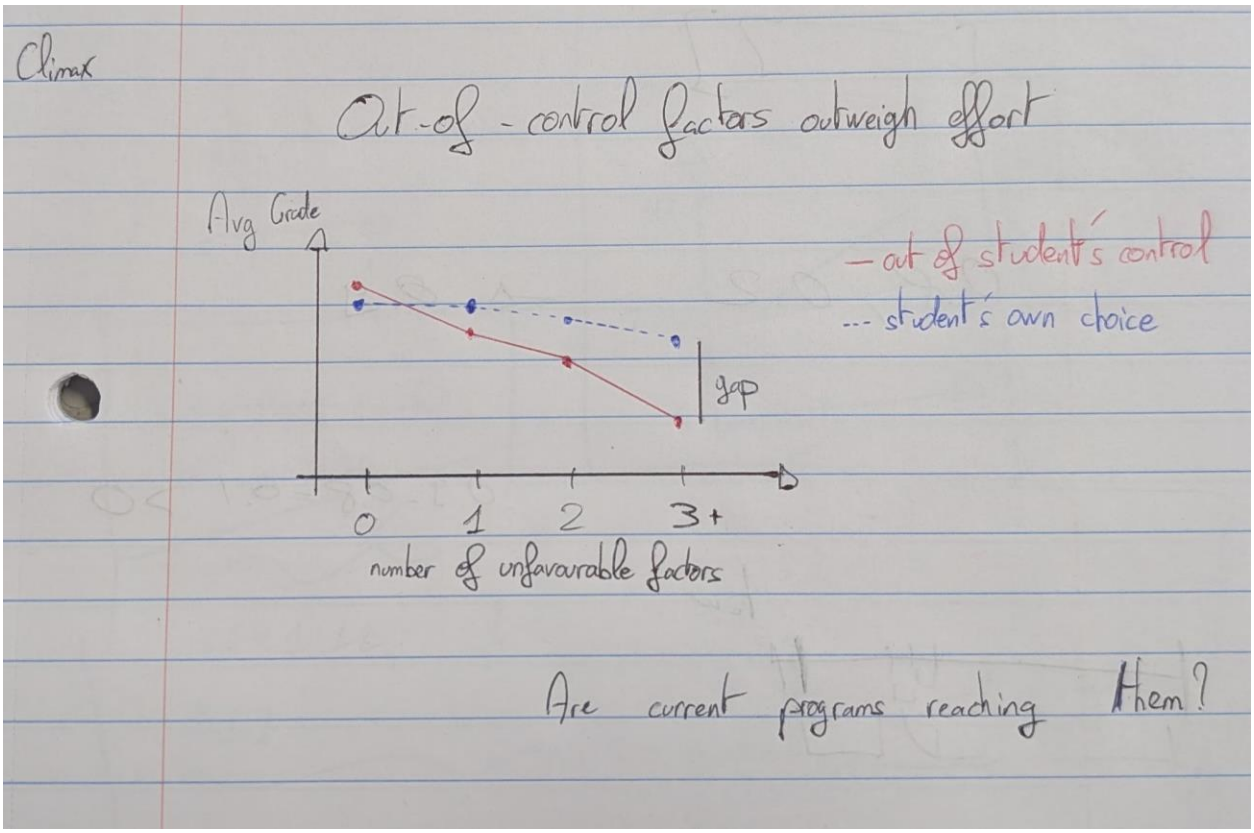
This sketch is intended for trising actions. This sketches reaches the following conclusion: as the level of disadvantage increases, the proportion of students receiving academic support decreases sharply, revealing a clear disconnect between needs and the measures implemented.

- Sketch ID 15 – Justin Bimal– Message ID 6:



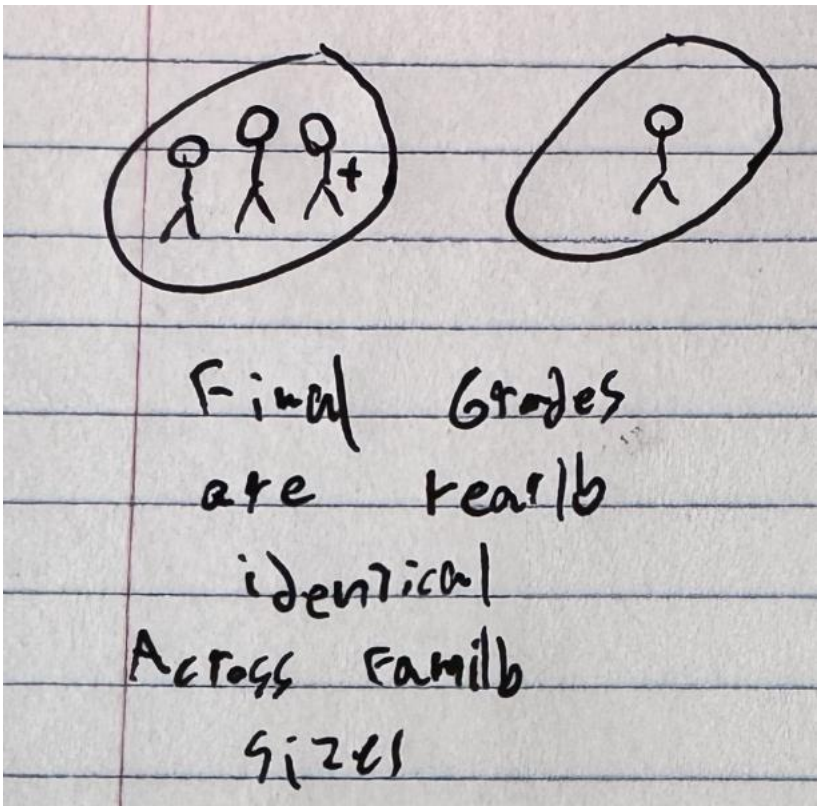
This sketch is intended for rising actions. This “small multiples” chart uses lines on a common dotted baseline to compare several factors, highlighting key variables with red markers. By plotting variables such as time spent studying, Internet access, previous failures, and commute time on identical vertical scales, observers can immediately see that only a few specific factors have a significant impact on students’ declining grades.

- Sketch ID 16 – Justin Bimal– Message ID 7:



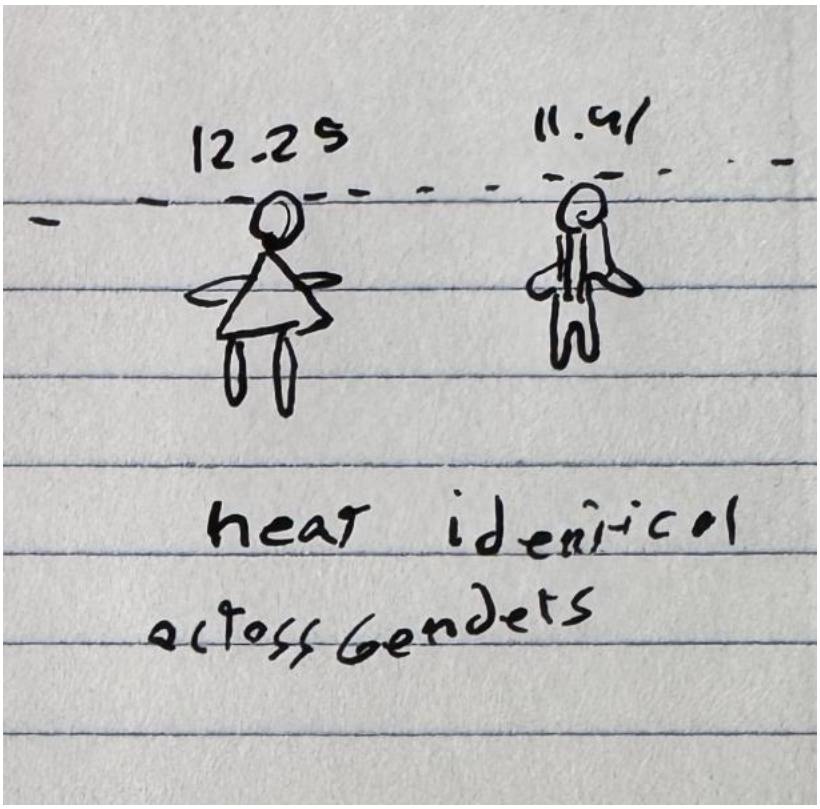
The widening gap between the two curves demonstrates that accumulating factors beyond student control leads to a much steeper decline in grades than personal habits.

Sketch ID 17 – Abdulaziz Alajlan



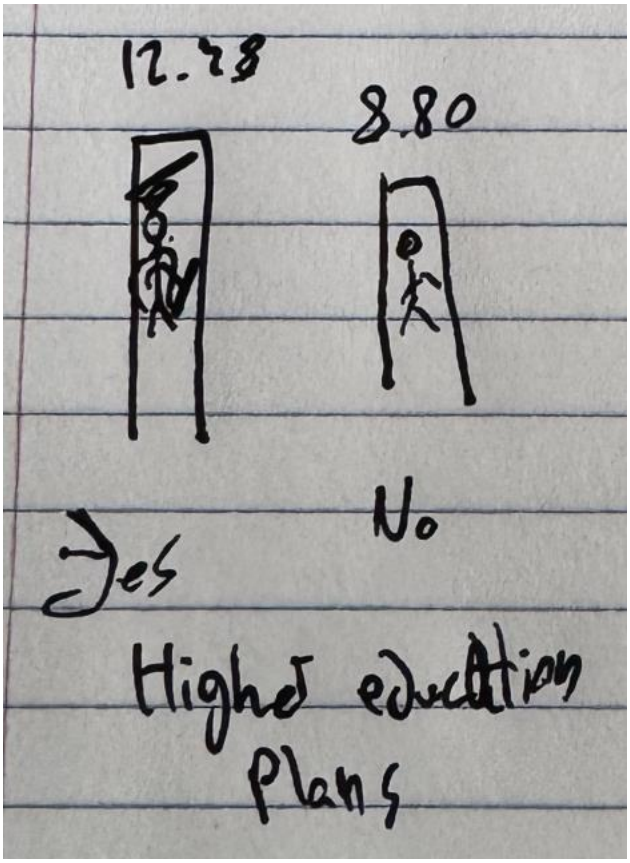
Pictogram: a family of three-plus and a single child side by side, with the final grade written under each. The near-identical numbers make the point that family size predicts nothing. Used for the hook.

Sketch ID 18 – Abdulaziz Alajlan



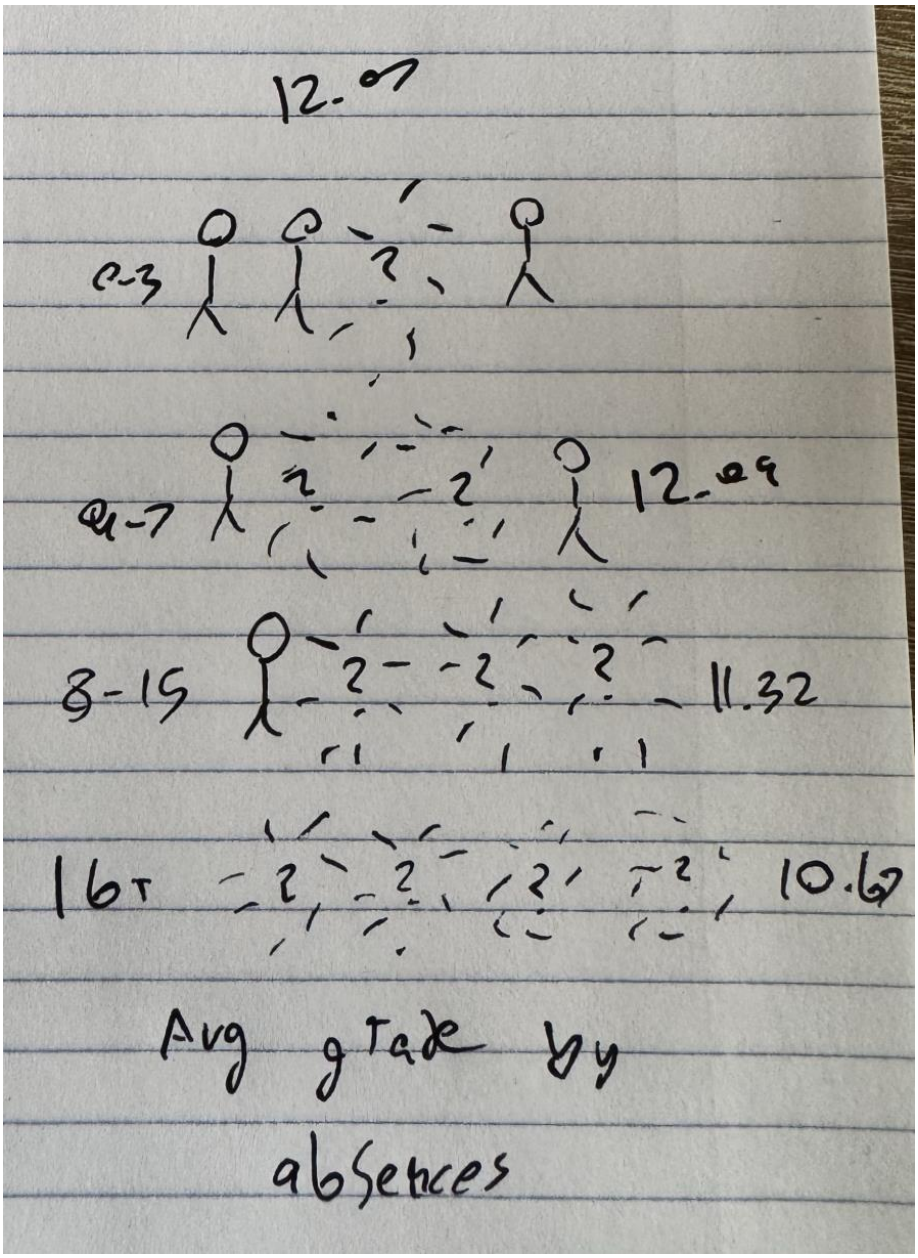
Two figures at heights set by mean final grade, 12.25 for female and 11.41 for male students, with a dashed line across the top showing how close they sit. Another demographic that does not separate students. Used for the hook.

Sketch ID 19 – Abdulaziz Alajlan



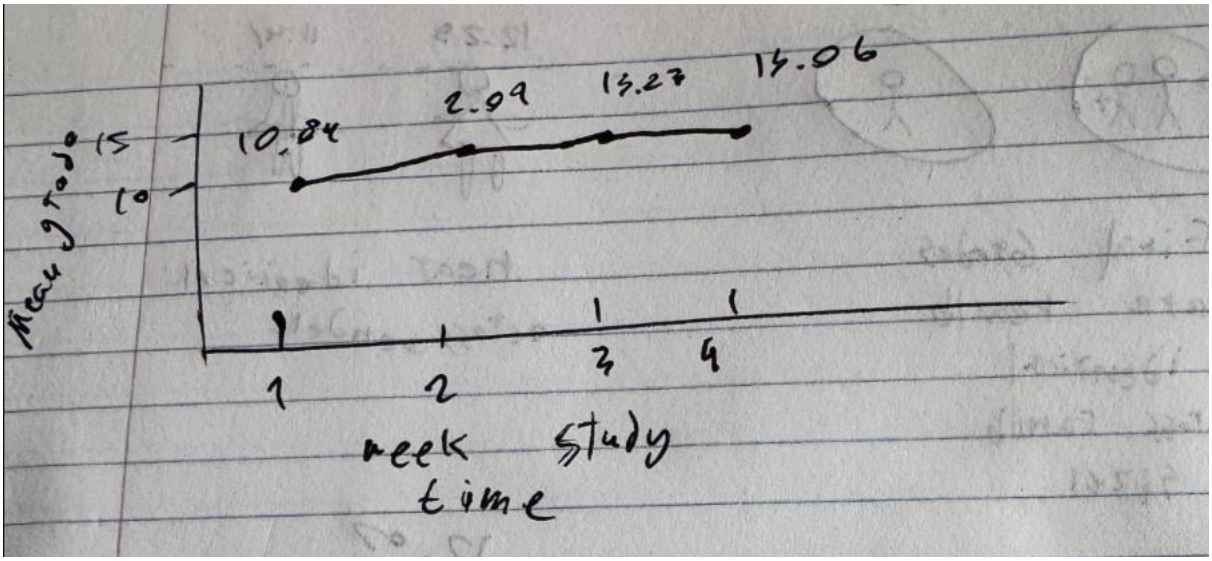
Two bars, each holding a student figure, with bar height encoding mean final grade for students who plan higher education versus those who do not: 12.28 against 8.80, three and a half points apart. Used for the rising insights.

Sketch ID 20 – Abdulaziz Alajlan



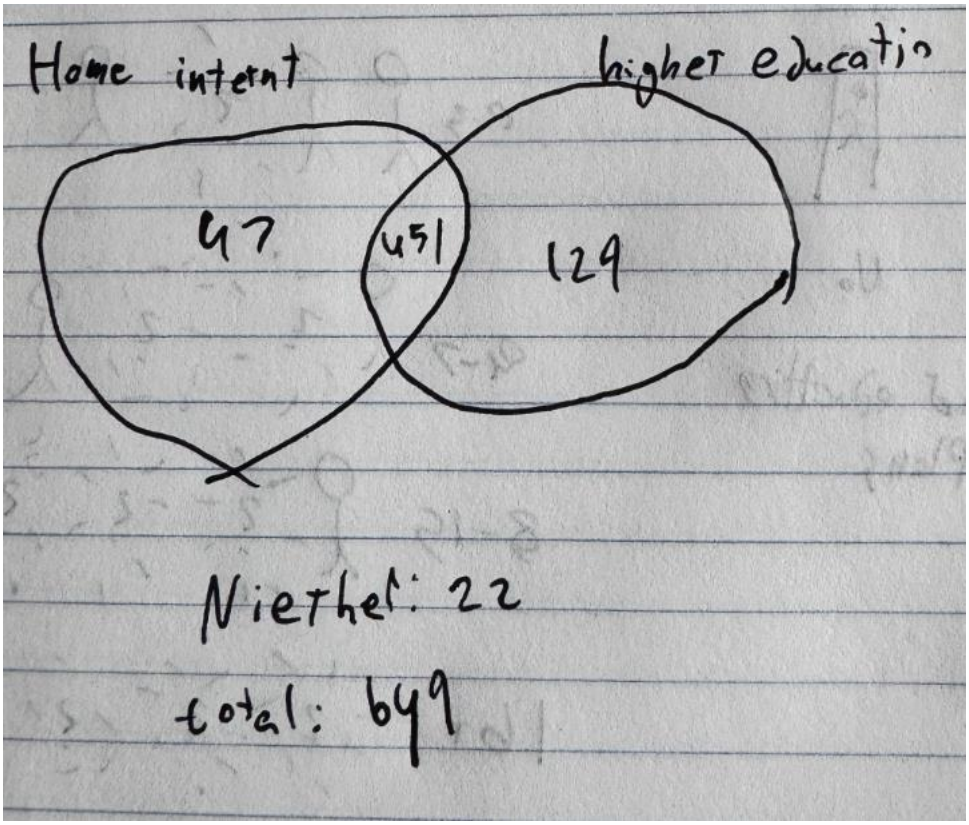
Unit chart with one row per absence band; present students are drawn solid and absences as faded "?" figures, and the mean grade sits at the end of each row, sliding from 12.07 and 12.04 through 11.32 down to 10.76. Used for the rising insights.

Sketch ID 21 – Abdulaziz Alajlan



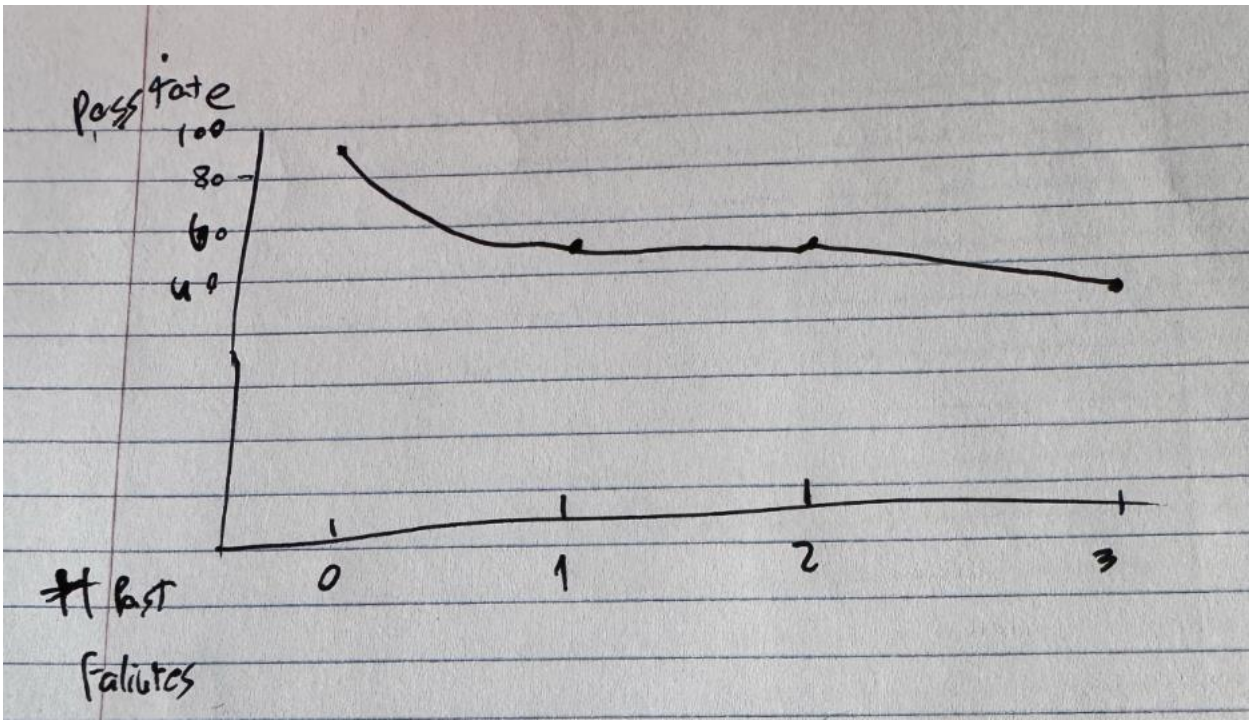
Line chart of mean final grade against weekly study time. The line climbs from 10.8 at under two hours to 13.2 at five to ten hours, then flattens to 13.1, so more study time stops paying off well before the top category. Used for the hook.

Sketch ID 22 – Abdulaziz Alajlan



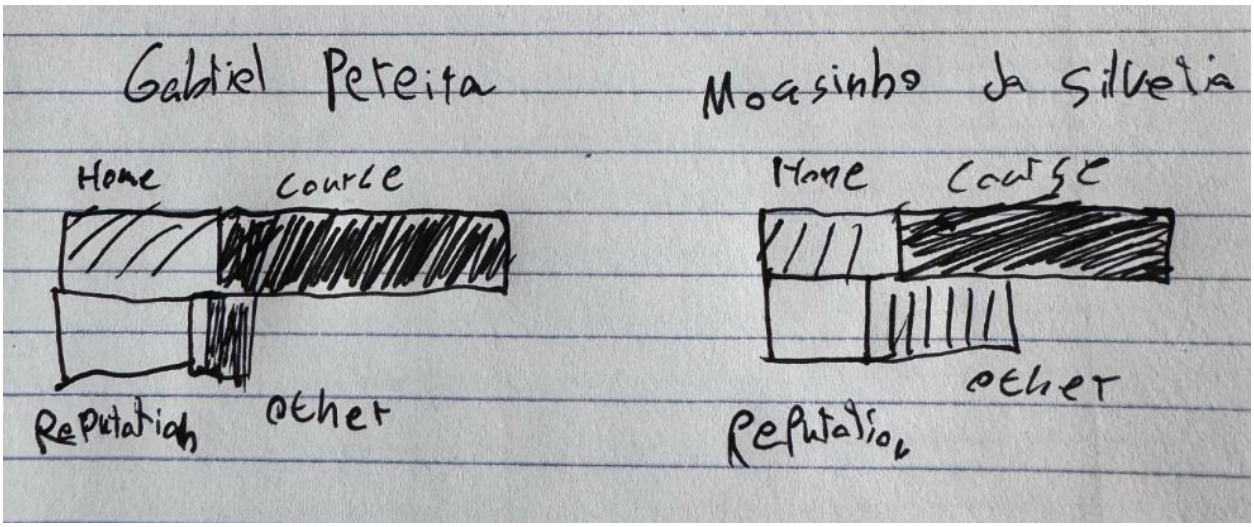
Venn diagram of home internet access and higher-education plans with counts in each region: 451 students have both, 47 only internet, 129 only plans, and 22 neither. The 22 with neither are a small, nameable group for outreach. Used for the solution.

Sketch ID 23 – Abdulaziz Alajlan



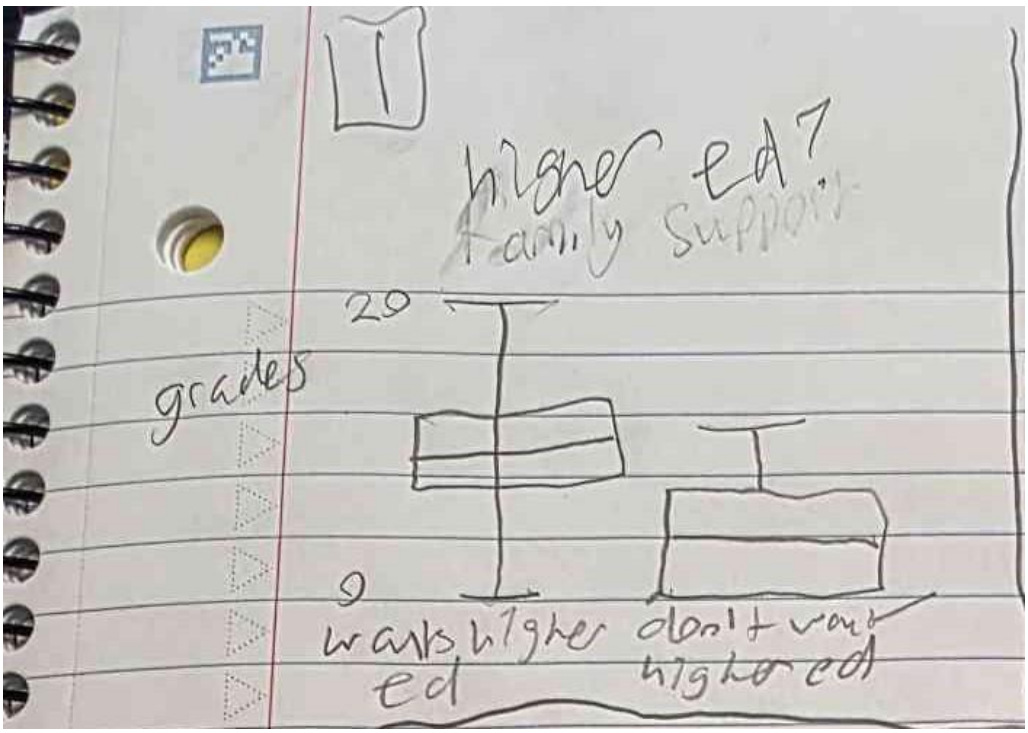
Line chart of pass rate against number of past failures, dropping from about 91% at zero to roughly 54%, 50%, and 36%. The steep first step is the main message. Used for the main message.

Sketch ID 24 – Abdulaziz Alajlan

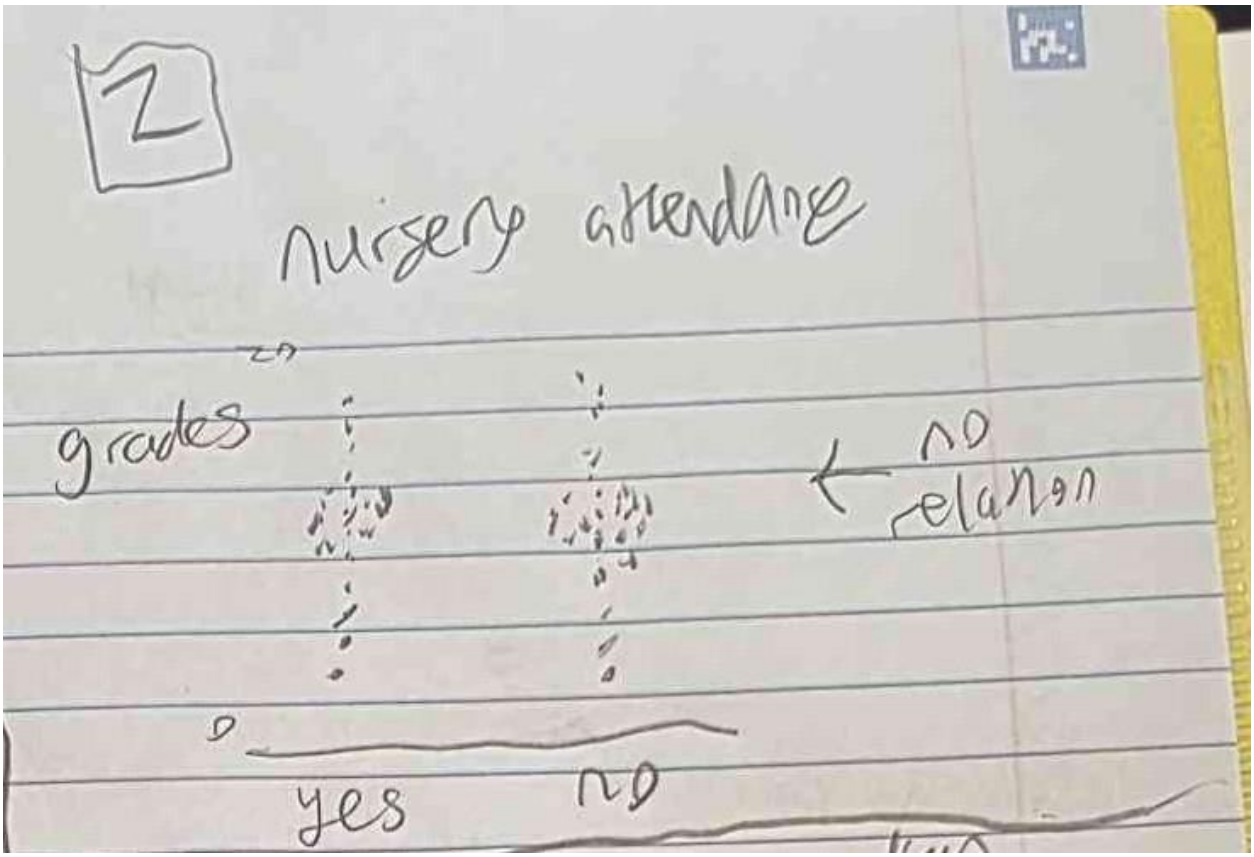


One mosaic per school showing the share of students who chose it for home, course, reputation, or other, so the two schools' different reason mixes can be compared at a glance. Used for the hook.

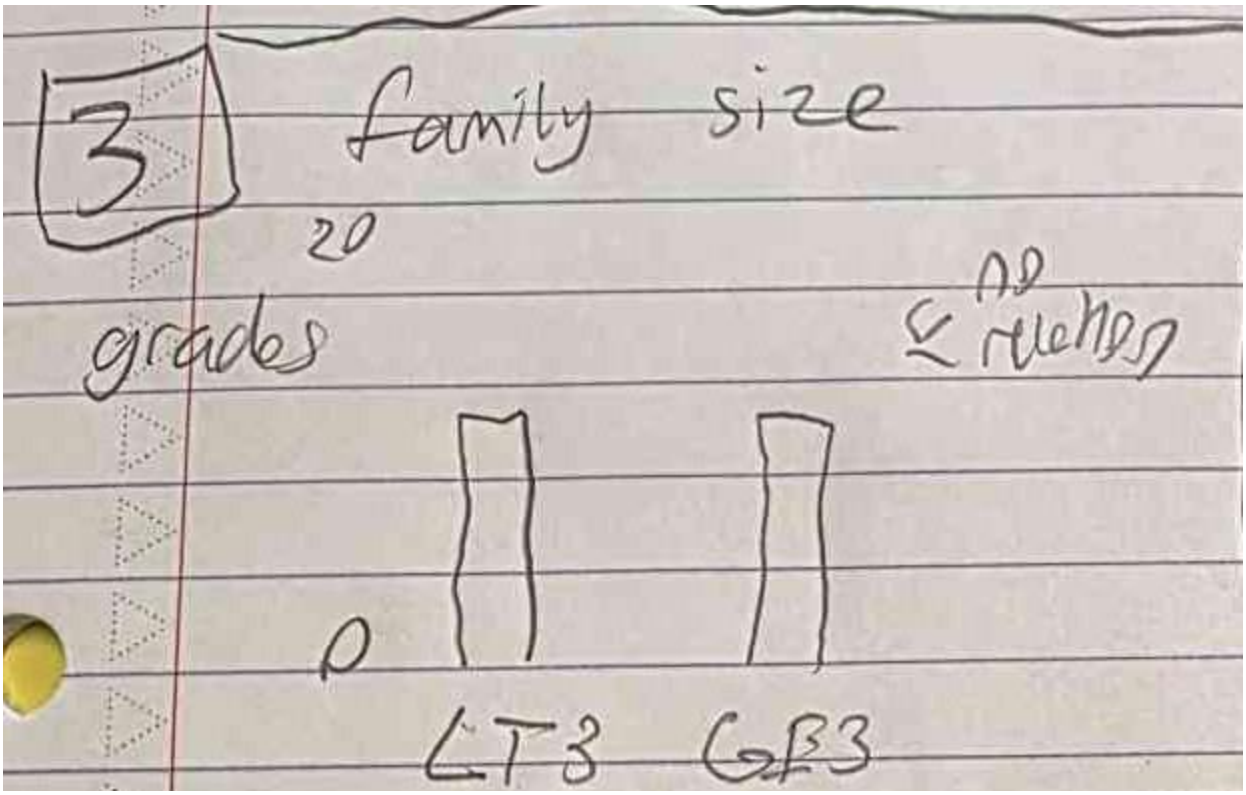
Sketch ID 25 – Joshua Blackmon – Message ID 9 – This sketch is for the rising action and shows that students who want to pursue higher education have overall better grades than those who don't.



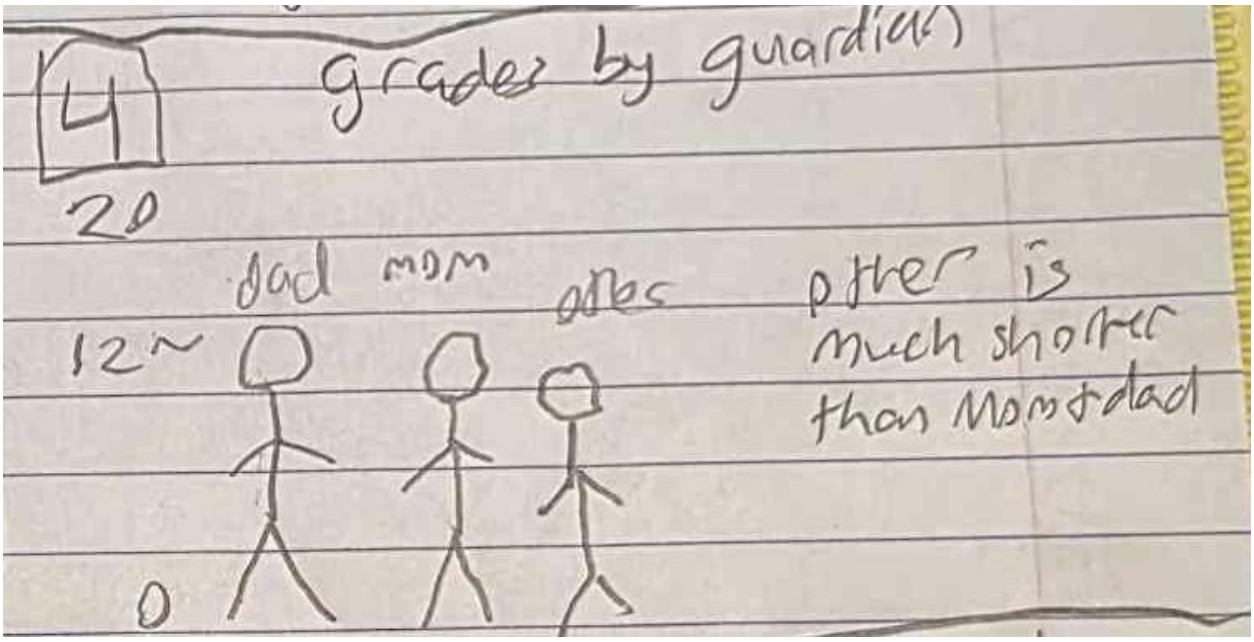
Sketch ID 26 – Joshua Blackmon – Message ID 10 – This sketch is intended for the hook and shows that things like nursery attendance have no correlation with grades.



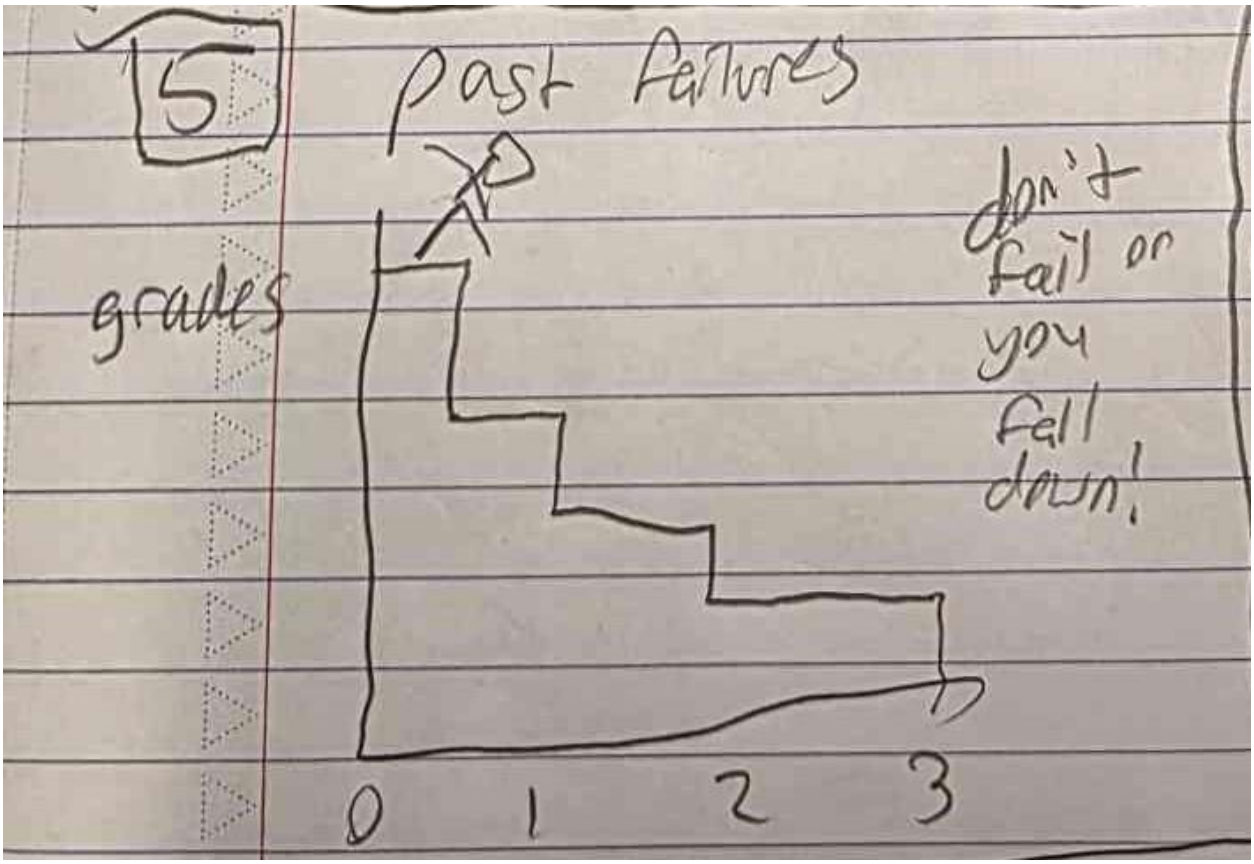
Sketch ID 27 – Joshua Blackmon – Message ID 10 – This sketch is also for the hook and shows that family size has basically no correlation with grades.



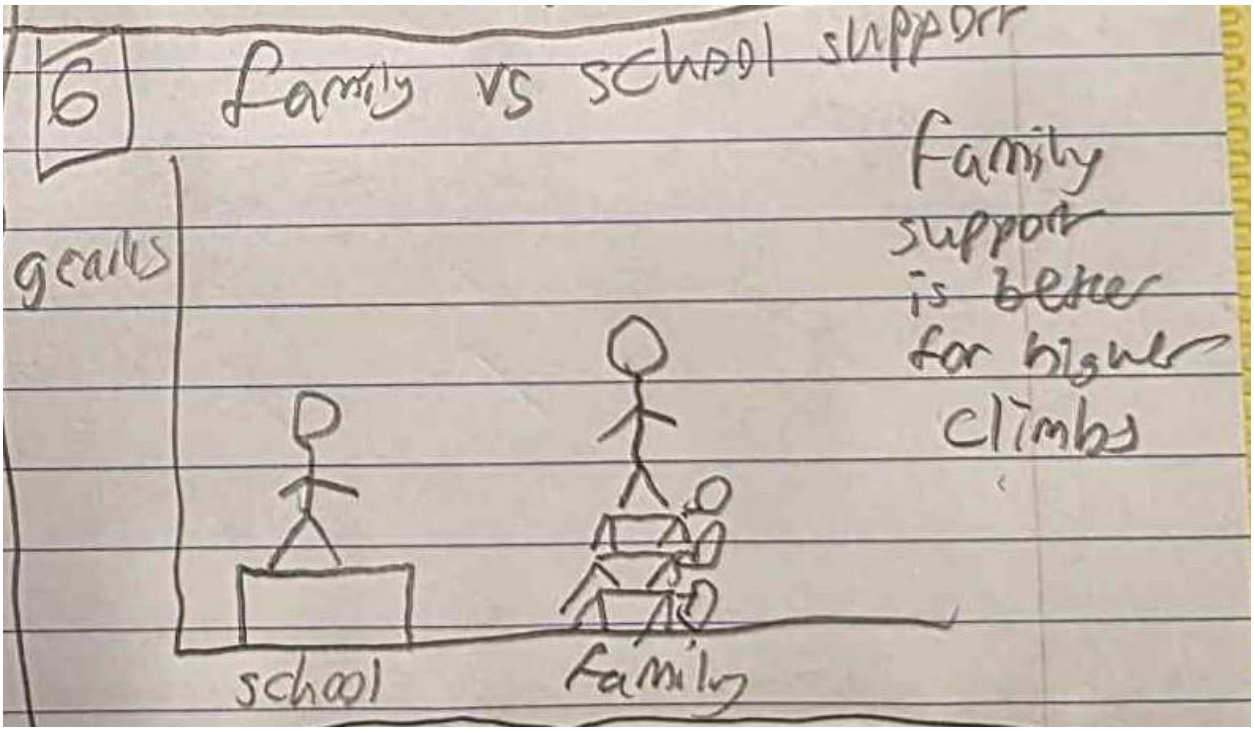
Sketch ID 28 – Joshua Blackmon – Message ID 5 – This sketch is intended for the rising action and shows that students with a non-parental guardian have lower grades overall.



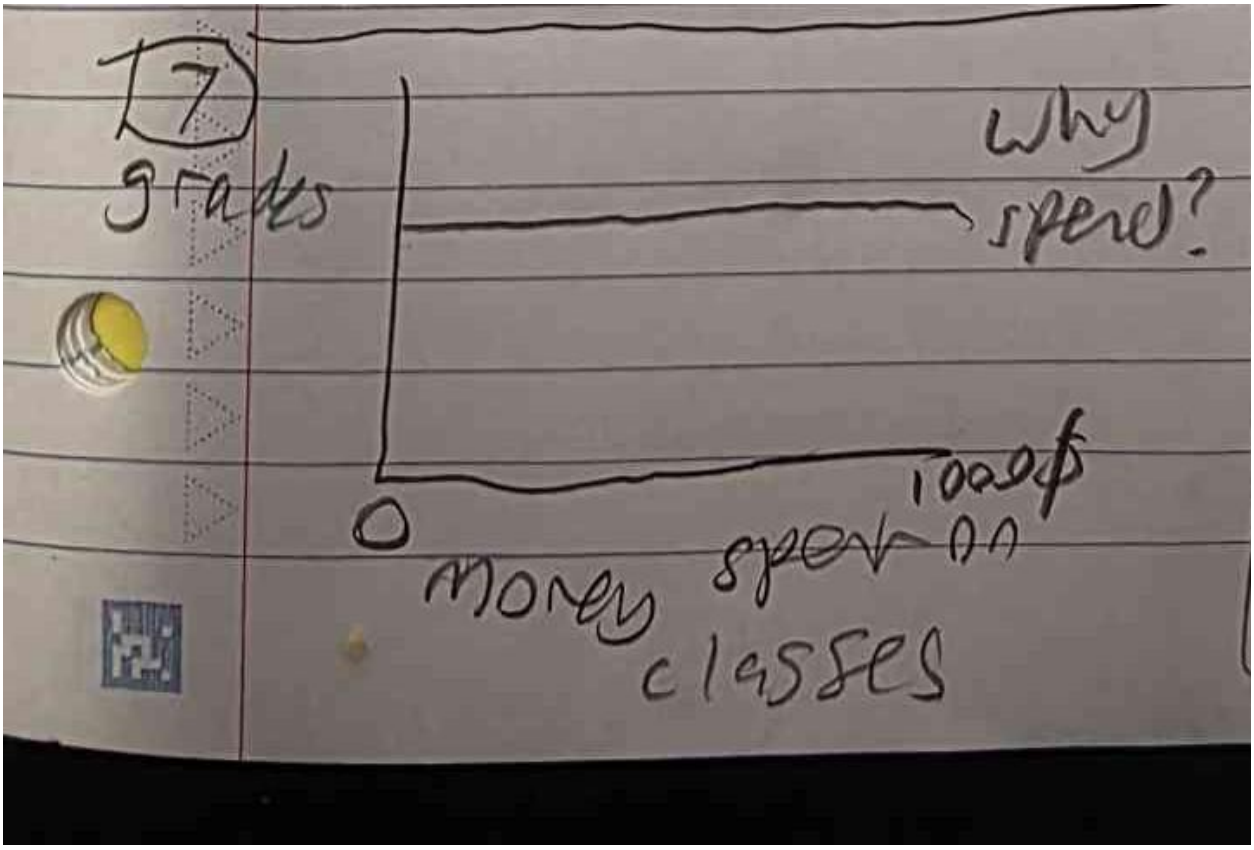
Sketch ID 29 – Joshua Blackmon – Message ID 11 – This sketch is intended for the main message and shows that getting a single failed class correlates with lowered average grades, lowering further per failed class.



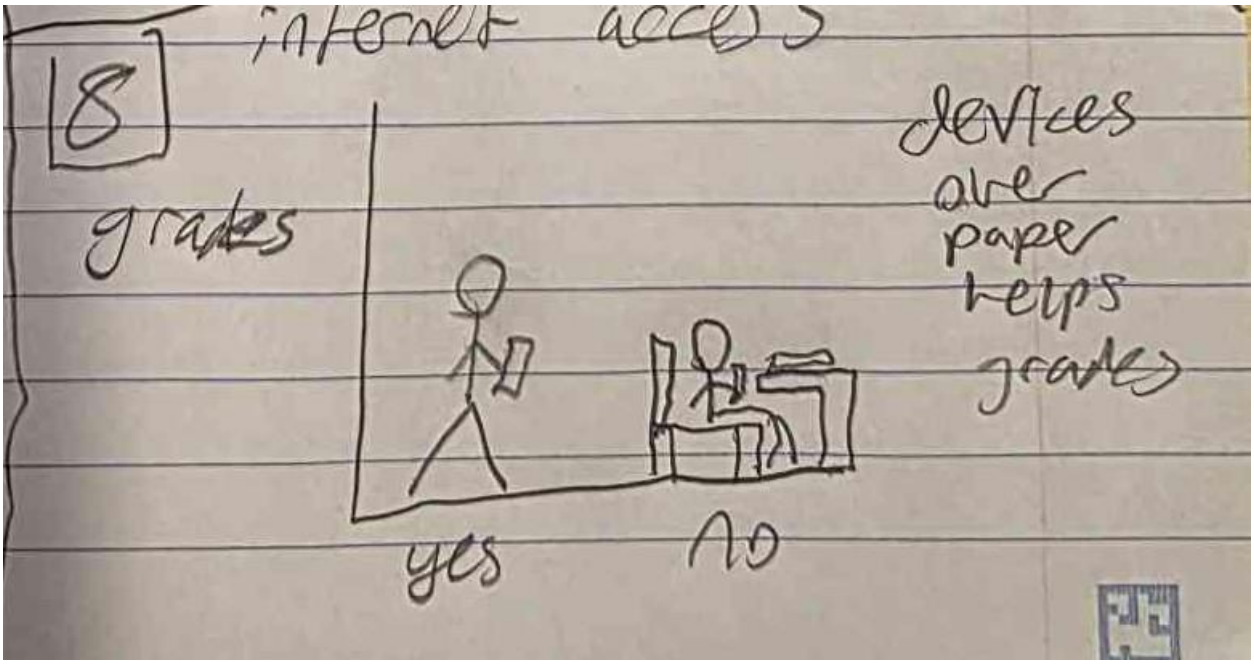
Sketch ID 30 – Joshua Blackmon – Message ID 8 – This sketch is intended for the call to action and shows that school support shows a correlation with lower grades when compared to students who receive family support.

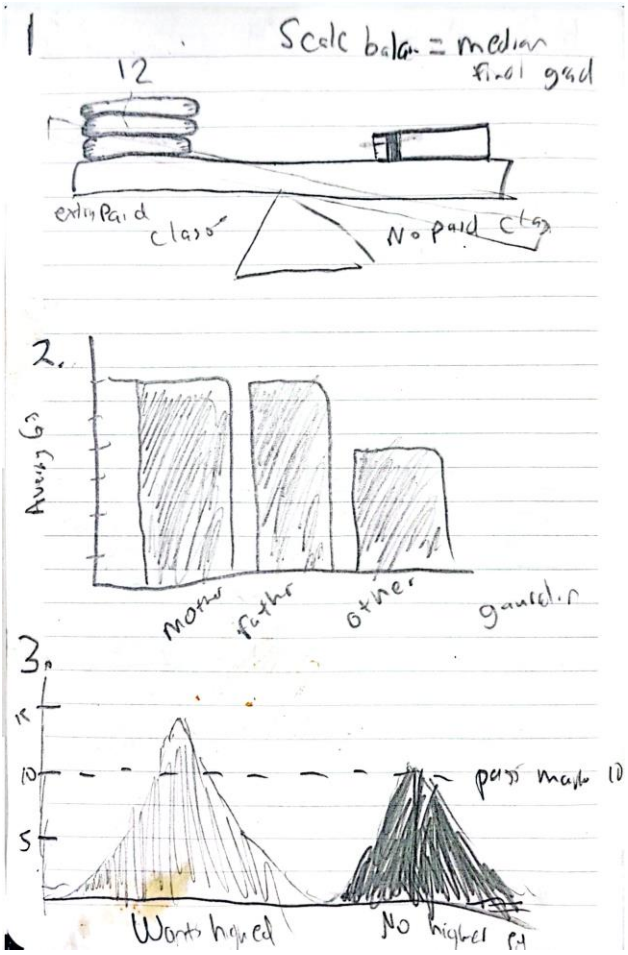


Sketch ID 31 – Joshua Blackmon – Message ID 10 – This sketch is intended for the hook and shows that grades are the same no matter how much money is spent on extra classes.



Sketch ID 32 – Joshua Blackmon – Message ID 5 – This sketch is intended for the rising action and shows that students with home internet access have higher scores overall than those that don't.

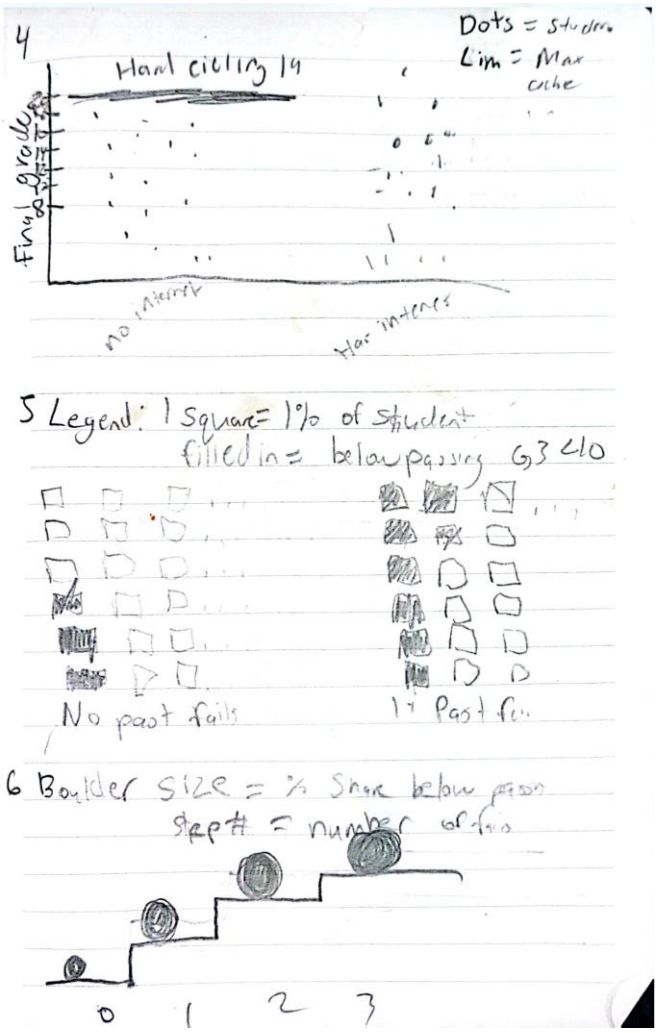




Sketch ID 33 – Timi Omoteso: I thought extra paid classes would give a huge advantage to students, but the visualization showed that is very much not the case. The median grade is exactly 12.0 whether a student pays for extra tutoring or not, so the effort pretty much just balances out.

Sketch ID 34 – Timi Omoteso: This visualization uses bar charts to look into guardian status to see its impact reveals a really noticeable drop-off in average grades (down to 10.5) for students who have a non-parent guardian, compared to those living with their moms or dads (around 12.4 to 12.5).

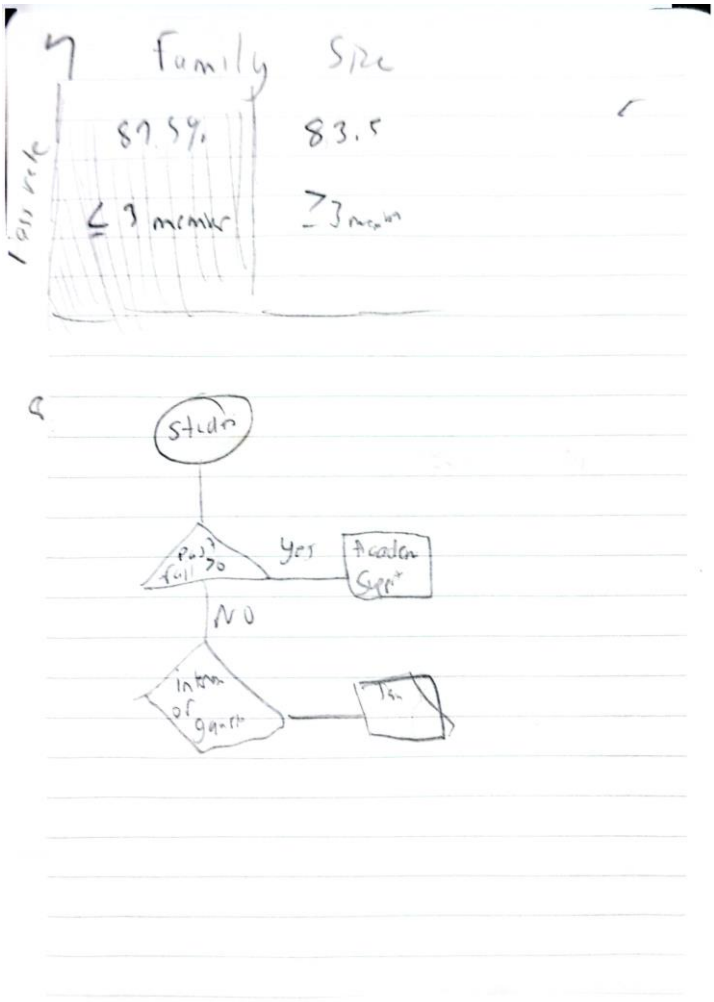
Sketch ID 35– Timi Omoteso This visualization looks at the spread of grades based on higher education goals shows a huge cluster of lower, failing scores for students who don't plan on going to college. This gives them a significantly lower baseline (median of 8.0) than the other students (median of 12.0).



Sketch ID 36 – Timi Omoteso: This is a scatter plot visualization investigating the maximum potential for these students shows that lacking home internet access acts like a hard ceiling. It caps those students at a max grade of 19, while the others can reach 20.

Sketch ID 37 – Timi Omoteso: This visualization helps show the trend of past failuers in relation to final grades. A waffle chart shows that almost half (49.0%) of the students with any past failure end up scoring below passing again, compared to only 9.3% of the students with a clean record.

Sketch ID 38 – Timi Omoteso: This visualization maps that failure risk scales using a step chart shows that with every past class failure a student has, their risk of failing the final grade goes up drastically. It steps from 9.3% up to 45.7%, 50.0%, and eventually hits 64.3%.

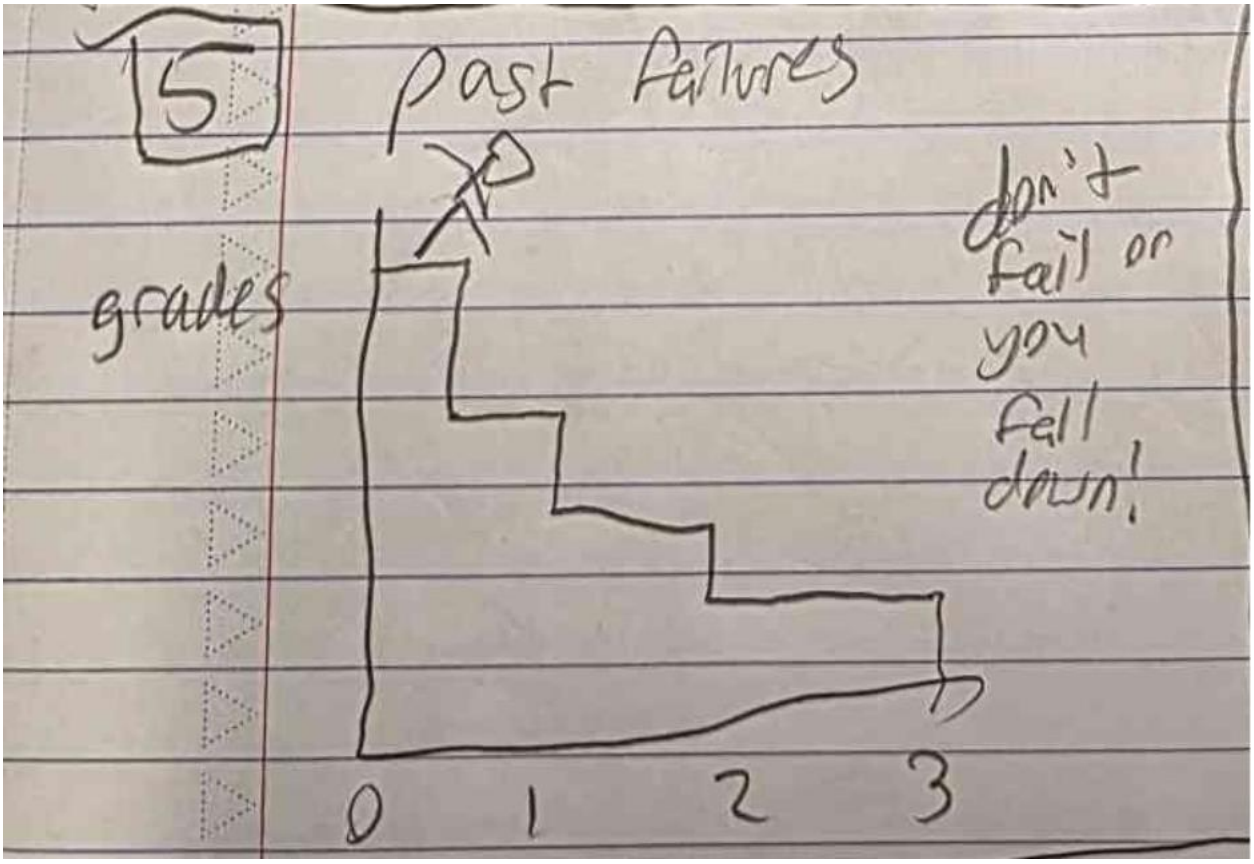


Sketch ID 39 – Timi Omoteso Lookibng into whether living in a larger family changed the risk of failing highlighted that family size is actually a very weak predictor of academic risk. Both groups pass at nearly identical rates (87.5% for families of 3 or fewer, and 83.4% for larger families).

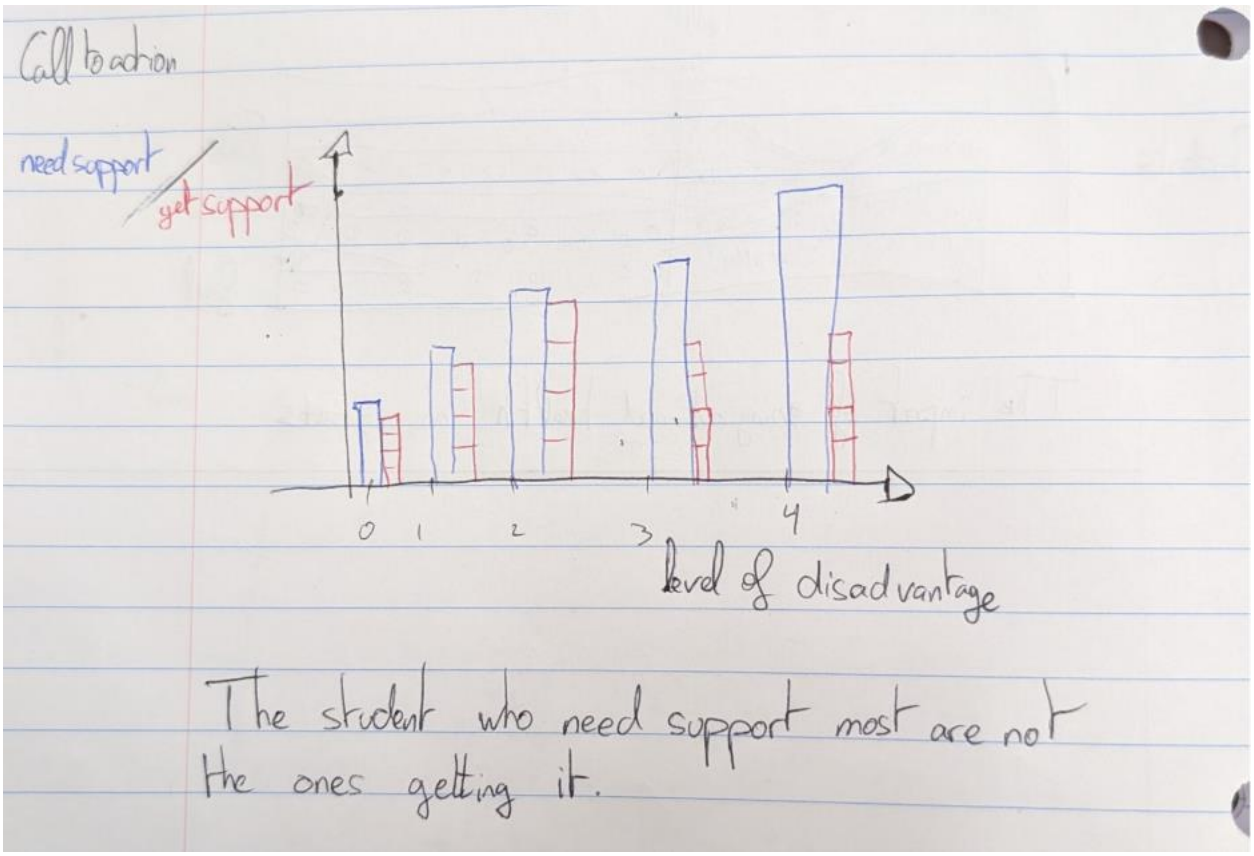
Sketch ID 40 – Timi Omoteso This decision tree maps out a clear policy for how a school board could use these findings. It shows that the school board should for sure use prior-failure records as the primary trigger for early outreach, followed by checking home context flags like internet or guardian status to provide targeted support.

Final Result:

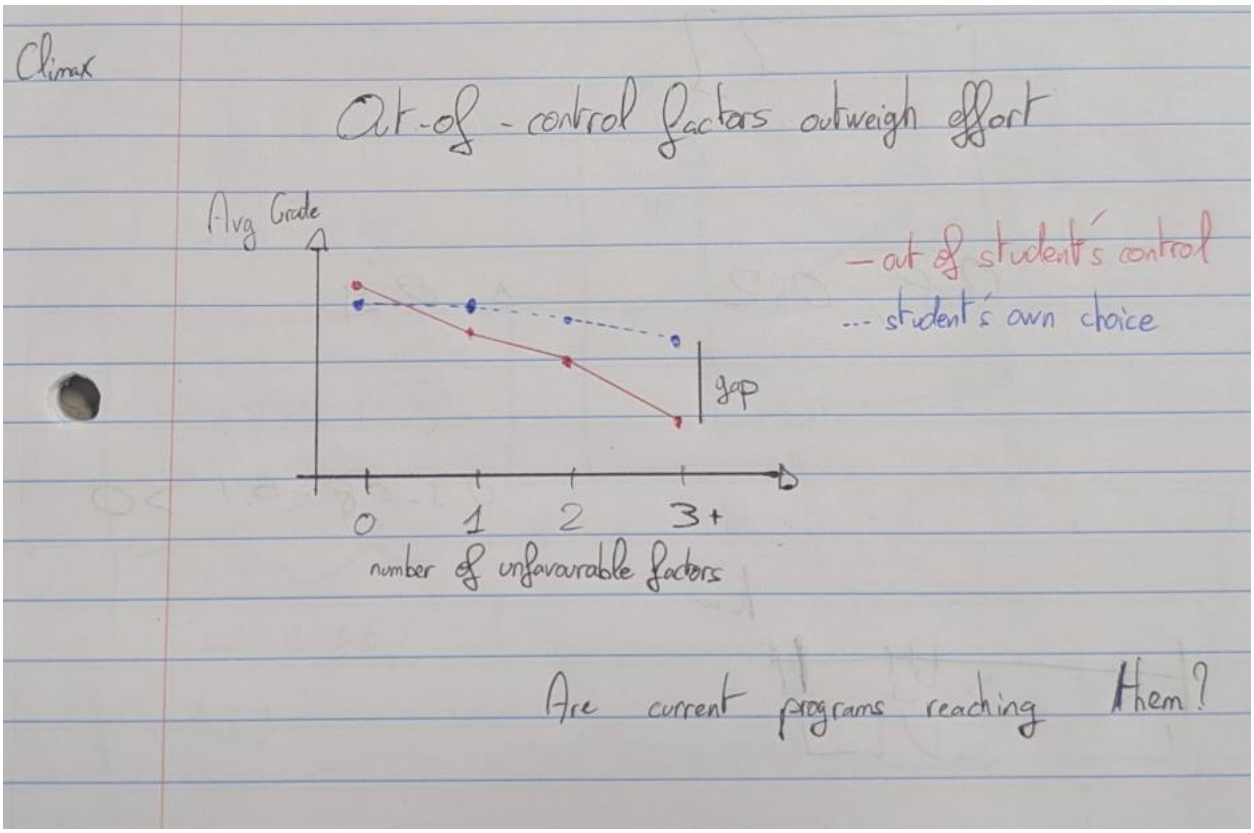
Q: How do grades correlate with past failures?



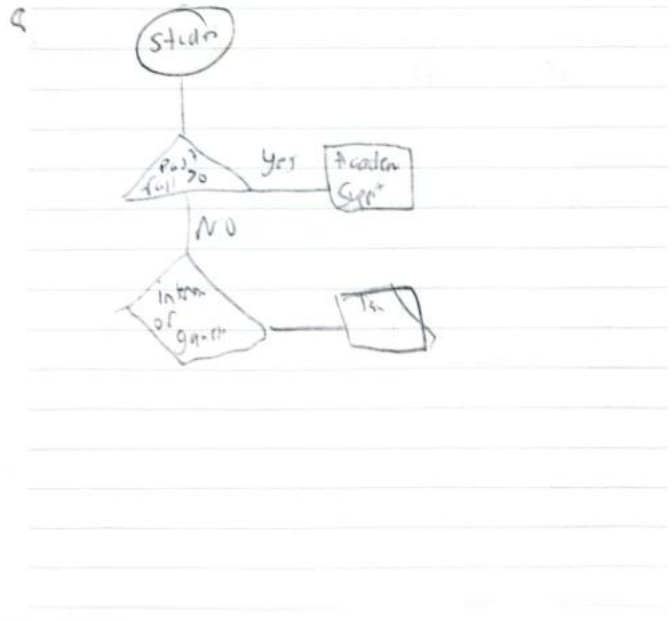
Q: Does the percentage of student receiving school supports increase as the number of disadvantages increase?



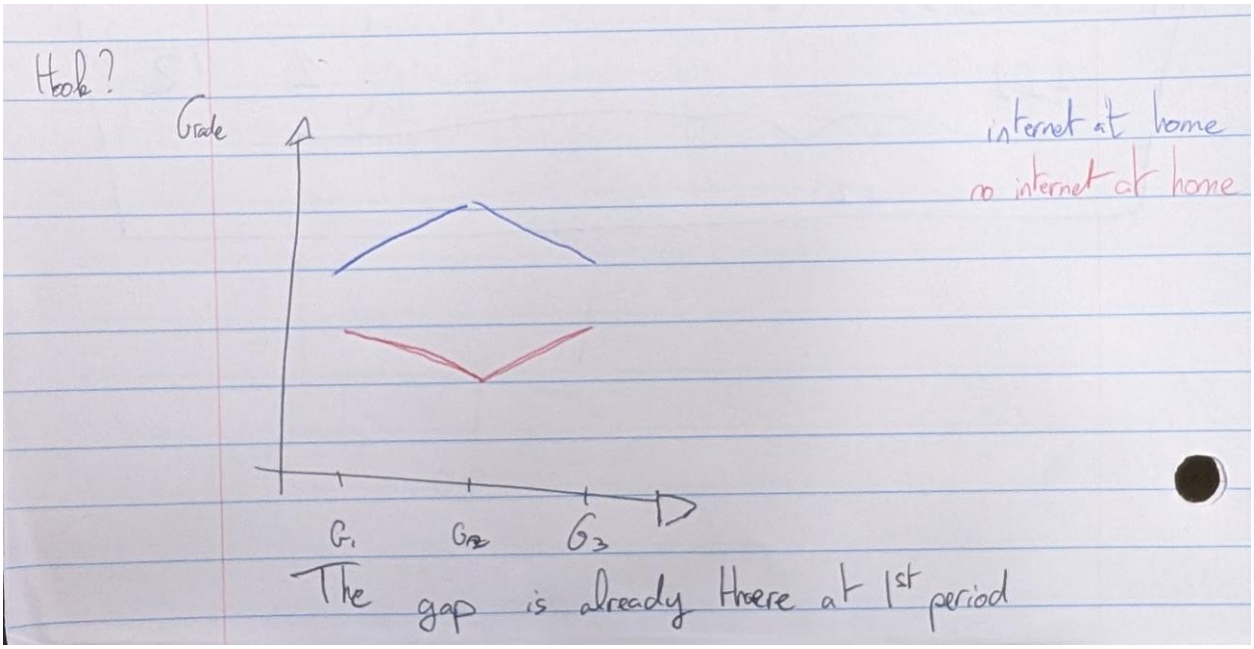
Q: Do grades decrease more as controllable disadvantages increase or uncontrollable disadvantages increase?



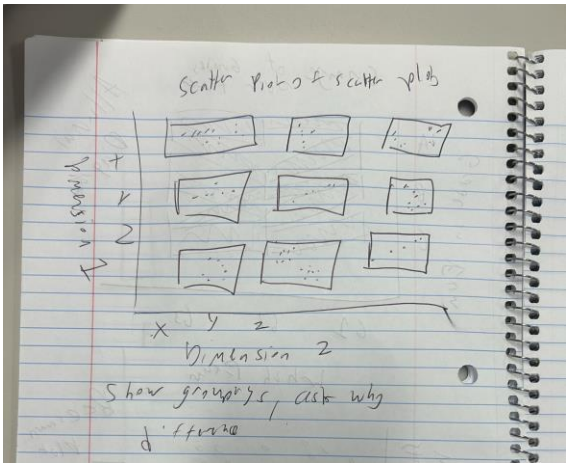
Q: What is the order in which factors should be considered when deciding on school intervention?



Q: Does the grade gap close at any point during the semester?



Q: Do some factors lead to different distribution of grades?



For most of the sketches, we chose them as basic examples of useful visualizations that could be expanded in our final project. We tried to select sketches that cover the entire story. Our focus was on unique and interesting visualizations because line and bar charts will be added to supplement them in the final product. The most relevant sketches are for use in the main message and call to action as core visualizations, while the later few are interesting ways to draw attention to our hook and rising action. Overall, we chose the

sketches that were most interesting (not simply a bar/line chart) and/or were very useful at specific points in our story. (Sketch 22 was not selected for this part because we talked to TA and they agreed that it would likely not be the best to implement).

Prototype Phase 1 Screenshots:

STUDENT PERFORMANCE EXPLORER

Understanding student achievement

Explore how family circumstances, access and study habits relate to grades in two Portuguese schools.

Observational data · 1,044 course records · Grades on a 0–20 scale

Subject: Both subjects | School: Both schools | Support comparison: School support | Reset filters

1044 records in this view

1,044	11.3 / 20	78.0%	79.2%
Course records	Mean final grade	Final grade ≥10	Internet at home

Explore the data

Hover or focus chart marks for values and sample sizes

01 / PREVIOUS FAILURES

The staircase of achievement

Mean final grades form the steps. A figure near the edge illustrates how precarious the next step can feel.

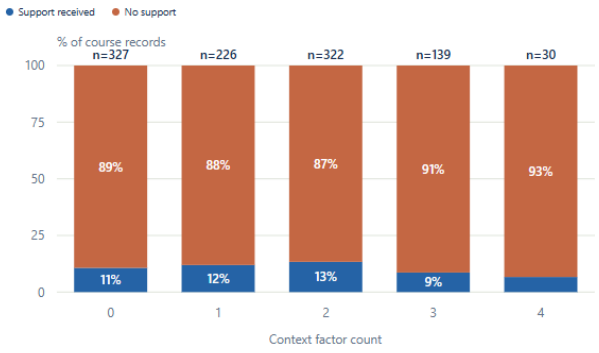


Step heights are observed group means. The person is an illustration; these groups do not track one student failing or predict an individual outcome.

02 / EDUCATIONAL SUPPORT

Who receives support?

Share of course records with and without support, by context factor count.

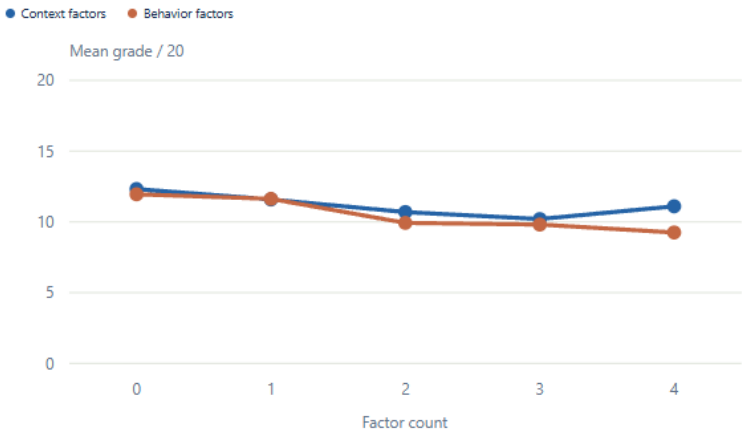


Support often goes to students already struggling. Differences cannot measure the effect of support.

03 / CONTEXT & BEHAVIOR

Two kinds of disadvantage

Mean final grade as each kind of factor accumulates.

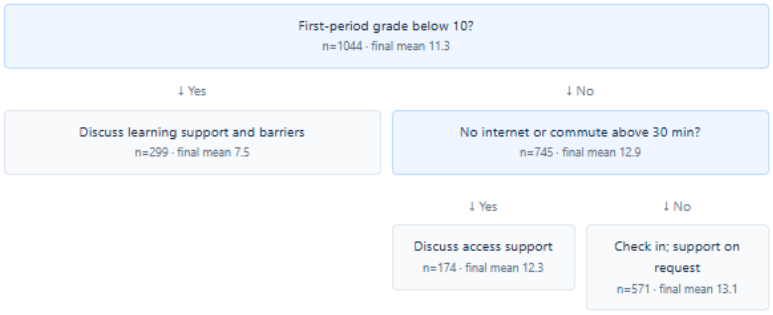


Each line groups the same records independently. Context and behavior are exploratory categories, not measures of personal responsibility.

04 / A SUPPORT CONVERSATION

Where could help begin?

An illustrative decision tree, with counts for the current filters.

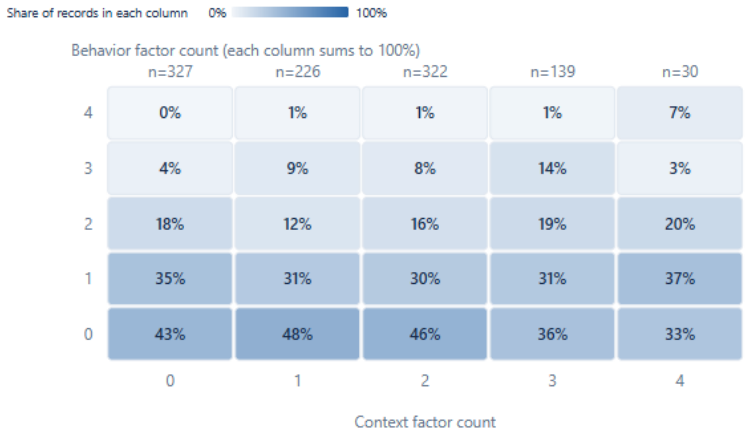


A transparent discussion aid, not a trained model or validated recommendation. Every student can request support.

05 / CONTEXT AND BEHAVIOR TOGETHER

Do the two kinds of disadvantage go together?

How records with each context factor count (columns) split across behavior factor counts (rows).



Each column sums to 100%: darker cells hold a larger share of that column's records. Column sample sizes (n) sit above the grid; hover or focus a cell for its count. These are exploratory counts, not validated scales; the pattern is an association and says nothing about causes or personal responsibility.

06 / INTERNET ACCESS

A year in three periods

Mean grade by internet access at home, from G1 through G3.



G1 and G2 are period grades; G3 is the final grade. Lines describe group means.



Data Link: https://github.com/yoshspider/Data-Vis-Principles-School-Success/blob/main/web/data.json

For phase 1, we chose to focus on the core website functionality, such as learning how to use filtering for the visuals. We created the core structure of a scrollable story and some visuals that could use further refinement and text. The visuals aren't organized as they will be in the final story, but they have accompanying text and filtering support to give interaction.

Prototype Phase 2 Screenshots:

CS 6730 · DATA VISUALIZATION · FALL 2026

From familiar assumptions to a concrete group

Four sections trace how the evidence narrows from weak demographic signals to past academic history.

Yash Agrawal · Justin Bimal · Joshua Blackmon · Oluwatimi Omoteso · Abdulaziz Atajjan

Mathematics and Portuguese · 1,044 course records (395 + 649) · Grades on a 0–20 scale

SCROLL TO BEGIN ↓

STORY

01 / HOOK

Familiar factors do not identify risk

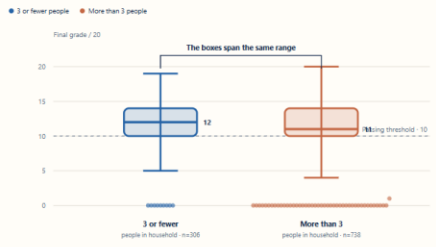
Three common assumptions about attendance, family size and paid classes do little to separate students by achievement.

Nursery attendance has only a small relationship with grades (about 0.4 points in mean final grade), family-size groups have medians one point apart and identical middle-50% ranges, and paid classes show no grade advantage (paid-class records average slightly lower). These weak results challenge the assumption that familiar demographic or financial factors automatically identify the students at most risk.

FAMILY SIZE · FINAL GRADE

Different household sizes, nearly the same grades

Final-grade distributions for households of three or fewer people and households of more than three.



Medians: 12 vs 11 Same middle 50%: 10–14

Boxes show the middle 50%; center lines mark medians; whiskers extend to the most distant values within 1.5 × IQR. Zero grades are retained. Mathematics and Portuguese course records combined, n=1,044; some students appear in both subjects.

02 / RISING INSIGHTS

02

Goals, access and home context matter more

Three indicators show clearer separation, shifting attention from broad demographics toward student circumstances.

Students planning to pursue higher education have higher grades (mean 11.6 vs 8.4), students with home internet average about one point higher (11.6 vs 10.5), and students whose guardian is neither parent average about 1.3 points lower than those with a mother as guardian. These findings narrow the story from broad assumptions to indicators connected with student goals, access, and home context.

HOME INTERNET

Internet access across three periods

Mean grade by internet access at home, from G1 through G3.



G1 and G2 are period grades; G3 is the final grade. Both subjects combined (n=1,044 course records); lines describe group means.

03 / MAIN MESSAGE

Past failure marks a concrete group for attention

Students with any recorded past failure are far more likely to fall below the passing threshold.

Among students with any past failure, 55.2% score below 10 and mean G3 is 7.99; among students with no past failure, 15.0% score below 10 and mean G3 is 12.05. Past academic history therefore provides the school board with a concrete group for early attention.

55.2%	7.99 / 20	15.0%	12.05 / 20
Below 10 · any past failure	Mean G3 · any past failure	Below 10 · no past failure	Mean G3 · no past failure

PREVIOUS FAILURES

The staircase of achievement

Mean final grade by recorded past failures. Step heights follow the observed group means.



Both subjects combined. n=1,044 course records (some students appear in both subjects). Groups with few records can produce unstable averages; the illustration is not a prediction about an individual student.

04 / SOLUTION

Start with prior-failure records

The recommendation uses existing records to target outreach, then tests whether the support actually helps.

Use prior-failure records to begin early outreach, assess whether internet access or guardian circumstances create additional needs, offer targeted academic support, and track later outcomes to evaluate whether the support works. The lower observed average among school-support recipients may reflect that support is assigned to students already struggling, not that support harms grades.

A SUPPORT CONVERSATION

Where could help begin?

An illustrative decision tree, with record counts and mean final grades.

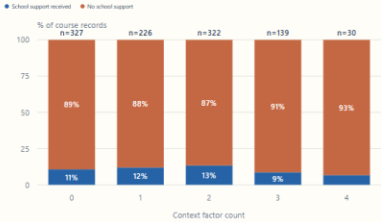


A transparent discussion aid, not a trained model or validated recommendation. Every student can request support.

WHO RECEIVES SUPPORT?

Support follows need

Share of course records with and without school support, by context factor count.



Support often goes to students already struggling, so differences cannot measure the benefit of support.

Advanced Visualization Design:

The node network shows how relatively correlated the negative factors are. A link that is thicker shows a higher rate of students with both traits than would be expected by the percentage of each negative alone.

06 / HOW DISADVANTAGES CONNECT

Which disadvantages travel together?

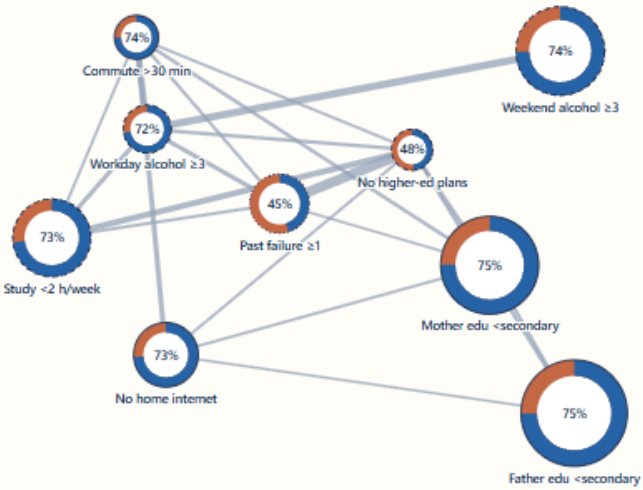
Each circle is a risk factor; a link joins two factors that occur together more often than chance. Click a circle or link to select those records in the matrix below; drag circles to rearrange.

Clear selection

Minimum lift 1.3×

1,044 course records · click a factor or link to select

Strongest tie: **No higher-ed plans + Past failure ≥1**, 2.9× chance · 37% pass vs 78% overall Select



Weekend alcohol ≥3

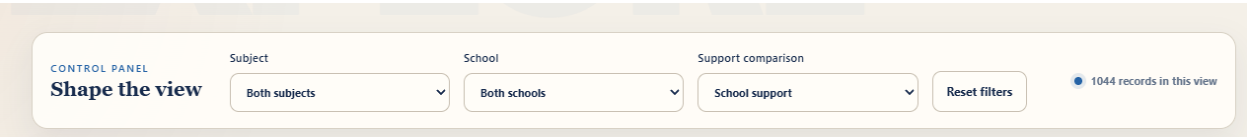
Records	411
Pass rate	74% vs 78%
Strongest tie	Workday alcohol ≥3 2.4×

Final grade ≥10 / below 10 · Context · Behavior · History · Link width = lift

Lift 2× = twice as often as chance (pairs with at least 10 shared records). An association, not a cause.



Above is our filtering/brushing visualization design. Clicking on a column (or multiple with the add to selection box) highlights the data points in other plots. This can be expanded or changed in the final product, but this is our current design for it.



This filtering system is our rough design for how filtering will work for our datasets, but we need to change it from a full site filter to each individual visual that it applies to. This will be our next step in the prototyping process.

Yash Agrawal · Justin Bimal · Joshua Blackmon · Oluwatimi Omoteso · Abdulaziz Alajlan

All of these students (the entire team) worked on the code and designs for this prototype (1 and 2)

Prototype Week 3 Screenshots:



01 / HOOK

Familiar factors do not identify risk

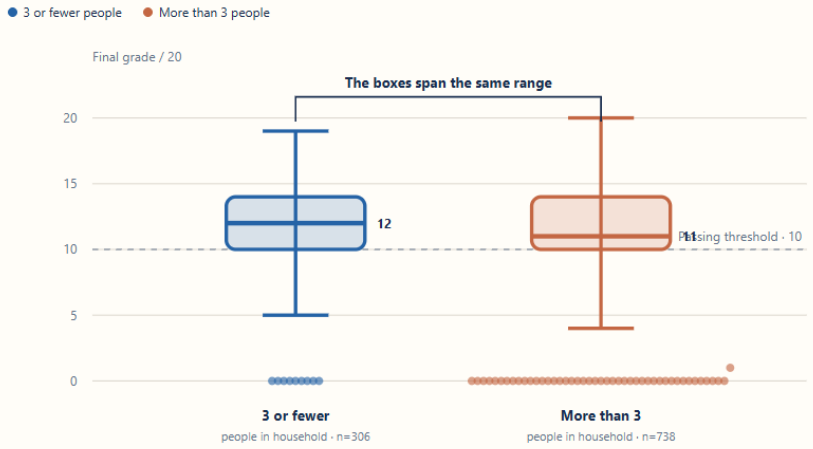
Three common assumptions about attendance, family size and paid classes do little to separate students by achievement.

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FAMILY SIZE · FINAL GRADE

Different household sizes, nearly the same grades

Final-grade distributions for households of three or fewer people and households of more than three.



Medians: 12 vs 11 Same middle 50%: 10–14

Boxes show the middle 50%; center lines mark medians; whiskers extend to the most distant values within 1.5 × IQR. Zero grades are retained. Mathematics and Portuguese course records combined, n=1,044; some students appear in both subjects.

HIGHER-EDUCATION PLANS

How grades flow from G1 to G3

Each ribbon follows course records from one grade band to another across G1, G2 and G3. Click the button (or the diagrams) to reveal G1, then G2, then G3. The top diagram covers students who plan higher education, the bottom one those who do not.

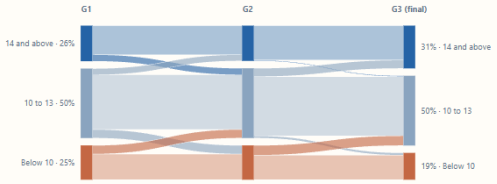
Motivation splits the grade distribution. Students with no higher-education plans tend to stop at a bare pass or below: 52% finish under 10, 47% land at 10 to 13 and only 1% reach 14 or above (n=89). Students who plan higher education are far more likely to reach the top band (31%) and fail far less often (19% vs 52%). Higher ambition goes with a very different grade distribution, an association rather than a proven cause: plans are self-reported and may themselves reflect earlier success or struggle.

Plans higher education: 19% below 10 at G3 No higher-education plans: 52%

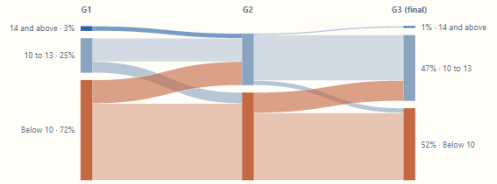
Start over ↺ Step 3 of 3 · G2 → G3 (final)

● 14 and above ● 10 to 13 ● Below 10

Plans higher education n=955



No higher-education plans n=89



G1 and G2 are period grades; G3 is the final grade. Bands: below 10 (failing), 10 to 13, 14 and above. Both subjects combined (n=1,044 course records); heights are shares within each group, so the two diagrams compare fairly even though the no-plans group is much smaller and its flows are less stable. Flows describe groups, not individual students, and show association, not cause.

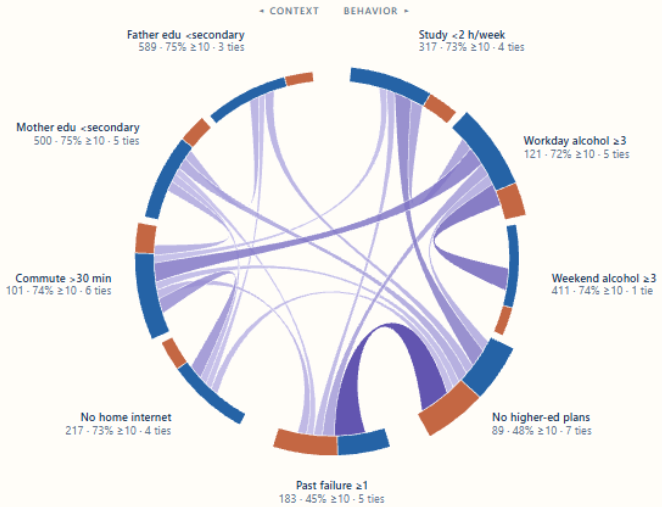
HOW DISADVANTAGES CONNECT

Which disadvantages travel together?

Each arc around the circle is a risk factor: context factors on the left, behavior factors on the right and past failure at the bottom. A ribbon joins two factors that occur together more often than chance. Click an arc or ribbon to select those records in the matrix below.

Clear selection | Minimum lift 1.3× | 1,044 course records · click an arc or ribbon to select

Strongest tie: **No higher-ed plans** + **Past failure ≥1**, 2.9× chance · 37% pass vs 78% overall Select · Most connected: **No higher-ed plans** (7 ties)



Workday alcohol ≥3

Records	**121**
Pass rate	**72%** vs 78%
Connection score	**4.5**

Ties shown, strongest first

Weekend alcohol ≥3	**2.4×**
Commute >30 min	**2.1×**
Study <2 h/week	**1.7×**
Past failure ≥1	**1.7×**

+ 1 more

Arc: final grade ≥10 / below 10

Thicker arc = more connected

Ribbon colour & width = lift (1× → 3×+)

Outer bar = share selected

Lift 2× = twice as often as chance (pairs with at least 10 shared records). Every factor gets an equal arc length; record counts appear beside each label. Arc thickness is the connection score: the sum of (lift − 1) over the ties shown. Ties between the same two groups are bundled together. An association, not a cause.

SCATTERPLOT MATRIX

What goes along with a higher final grade?

Three behaviors a school can act on: weekly study time, workday alcohol use and absences. Each cell compares the column variable (horizontal axis) with the row variable (vertical axis); the diagonal shows each variable's distribution. Every dot and bar is coloured by final grade, so look for where blue gives way to orange. Drag a box in any cell, or click a bar, to select records: the same records keep their colour in every other cell, and everything else turns grey. Selections made in the chord diagram above appear here too. Hold Shift (or tick "Add to selection") to combine several boxes and bars. The panel beside the matrix shows the final-grade distribution of whatever you select, set against all records.

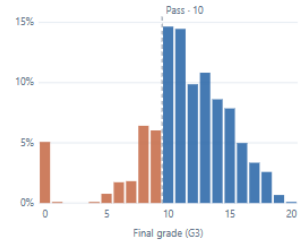
Clear selection ☐ Add to selection (or hold Shift) ☐ Only failed grades (final grade below 10)

1,044 course records · drag a box in any cell or click a bar to select



Final grade distribution

All 1,044 · mean 11.3 · 22% below 10 · 28% at 14+



03 / MAIN MESSAGE

Past failure marks a concrete group for attention

Students with any recorded past failure are far more likely to fall below the passing threshold.

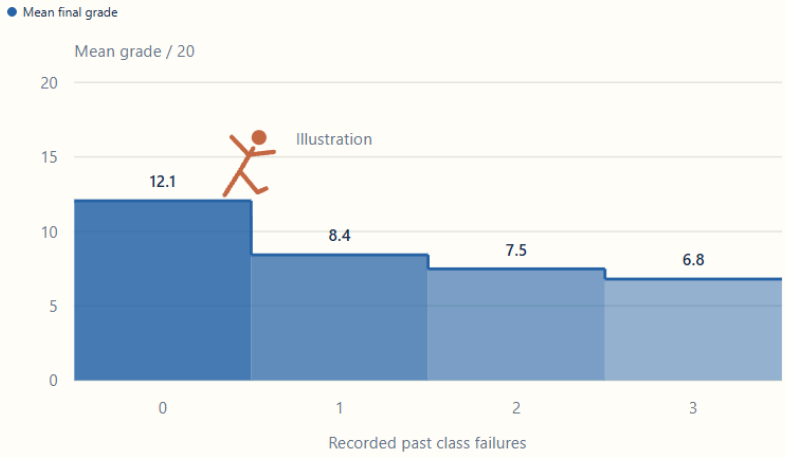
Among students with any past failure, 55.2% score below 10 and mean G3 is 7.99; among students with no past failure, 15.0% score below 10 and mean G3 is 12.05. Past academic history therefore provides the school board with a concrete group for early attention.

55.2%	7.99 / 20	15.0%	12.05 / 20
Below 10 · any past failure	Mean G3 · any past failure	Below 10 · no past failure	Mean G3 · no past failure

PREVIOUS FAILURES

The staircase of achievement

Mean final grade by recorded past failures. Step heights follow the observed group means.



Both subjects combined, n=1,044 course records (some students appear in both subjects). Groups with few records can produce unstable averages; the illustration is not a prediction about an individual student.

04 / SOLUTION

Start with prior-failure records

The recommendation uses existing records to target outreach, then tests whether the support actually helps.

Use prior-failure records to begin early outreach, assess whether internet access or guardian circumstances create additional needs, offer targeted academic support, and track later outcomes to evaluate whether the support works. The lower observed average among school-support recipients may reflect that support is assigned to students already struggling, not that support harms grades.

A SUPPORT CONVERSATION

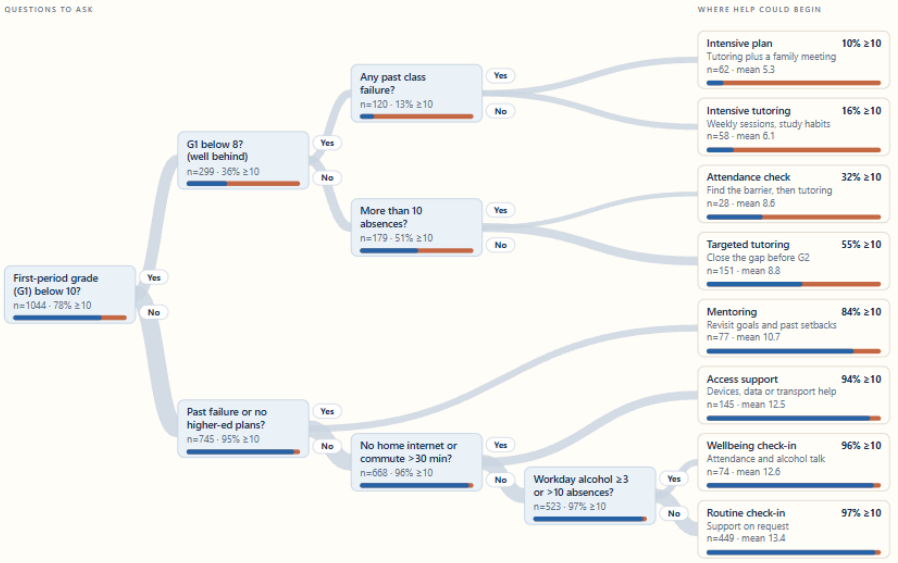
Where could help begin?

An illustrative decision tree: follow the questions from left to right to reach one of eight starting points, each with record counts, mean final grades and pass shares. Answer Yes or No beside each question to reveal the next one, or tick "Show full tree" to see every path.

Show full tree | Start over

Showing all 8 starting points. Point at any box or branch to trace its paths.

Share reaching a final grade ≥10 (blue) or not (orange) | Branch width = records on that path



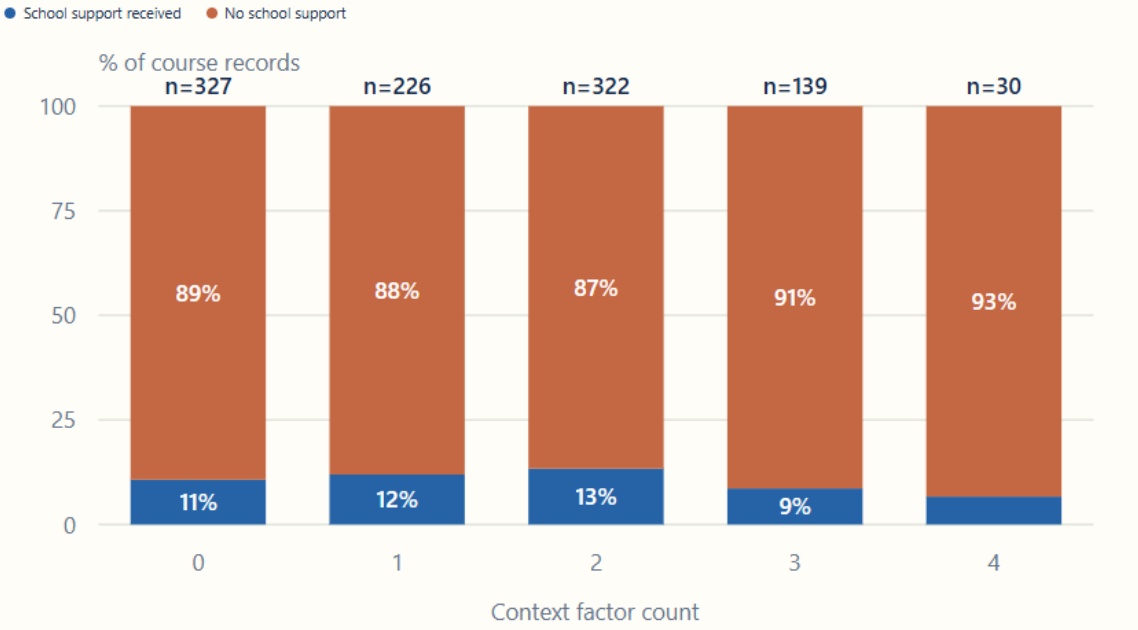
A transparent discussion aid, not a trained model or validated recommendation. Every student can request support.

WHO RECEIVES SUPPORT?

School support does not follow disadvantage

School support | Family support

School support is **not offered at higher rates to students with more disadvantages**. 11% of records with 0 or 1 context factors receive it, compared with 8% of records with 3 or 4. The share is 11% at zero factors, peaks at 13% with 2 factors and falls to 7% at four, so the students facing the most barriers are no more likely to be reached.



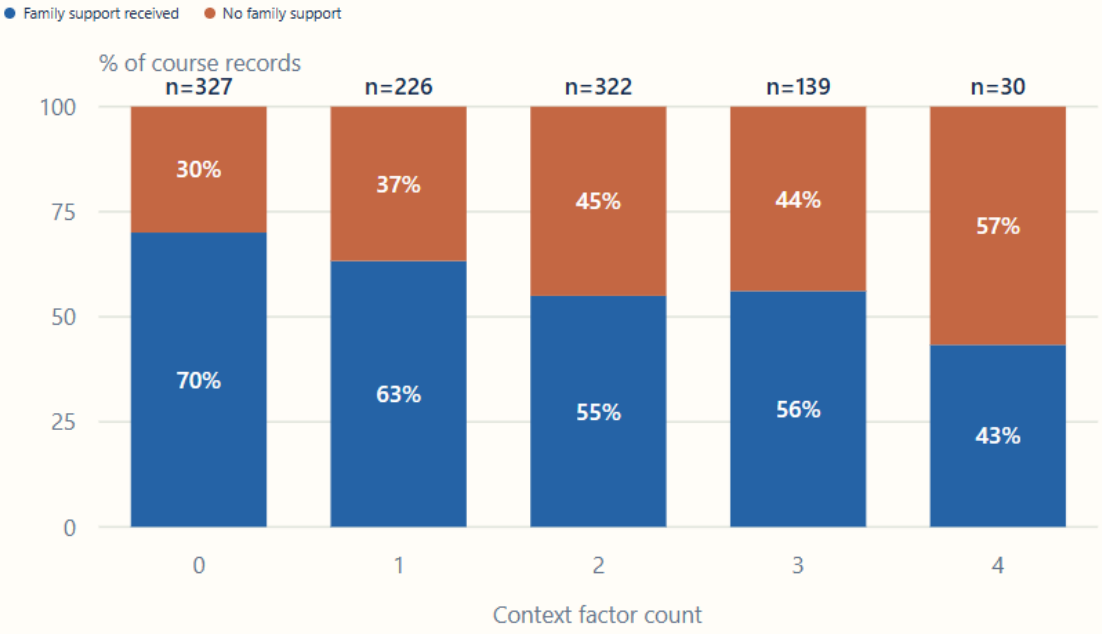
School support is a yes/no record, not a measure of how much help was given. Groups with three or four factors are small, so their shares are unstable. Differences cannot measure the benefit of support.

WHO RECEIVES SUPPORT?

Family support is highest with the fewest disadvantages

School support | **Family support**

Family support is **higher where the disadvantage count is low**. 67% of records with 0 or 1 context factors receive it, compared with 54% of records with 3 or 4. The share falls from 70% at zero factors to 43% at four, so families facing the most barriers are the least likely to report extra help at home.



Family support is a yes/no record, not a measure of how much help was given. Groups with three or four factors are small, so their shares are unstable. This is an association, not a cause.